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Effective interventions to mitigate psychological consequences from COVID-19 pandemic: a systematic review

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Title: Effective interventions to mitigate psychological consequences from COVID-19 pandemic: a systematic review

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ABSTRACT

Objective

This review assesses interventions and their effectiveness in mitigating psychological consequences from pandemic.

Method

Published English literature were searched from four databases [Medline, PubMed, Embase, and Psycinfo], Total 27 papers with 29 studies (one paper reported three studies) met inclusion criteria. Cochrane risk of bias tool is applied to assess quality of all randomized control trials (RCT).

Results

All studies were recently conducted in 2020. Publications were from by high- (13), upper-middle (12) and lower-middle income countries (3) and one global; 8 studies provided interventions to COVID-19 patients, six to health care workers and the remaining for general population. Of 29 studies, 14 were RCT.

All 14 RCT were assessed for risk of biases; five studies had low risk as measured against all six dimensions reflecting high quality study.

Of these 29 studies, 26 tools were applied; eight for anxiety, seven for depression, five for stress, five for insomnia and one for suicide. Tools used to assess the baseline and outcomes of interventions were standardized and widely applied by other studies with high level of reliability and validity. Of total 57 results mesured by 29 studies, 45 (78.9%) showed significant positive results in reducing psychological consequences; 11 out of 14 depression; 20 out of 24 anxiety; seven out of nine insomnia; seven out of nine stress showed positive results respectively. One intervention against suicide also showed mixed outcomes.

Conclusions

A majority, 78.9% of results from 29 interventions shows effective in mitigating psychological consequences from COVID-19 pandemic in health care workers, patients and general population.

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These effective interventions can be applied in other country settings, for which modes of delivery can be adapted in line with country resources, pandemic and health systems context.

Article summary

Strengths and limitations of this study

- This systematic review provides timely evidence on which interventions are effective in mitigating psychological consequences on populations and healthcare workers during the COVID-19 pandemic.
- Effective interventions from this review can be adapted and scaled up in population and healthcare workers.
- The heterogeneity of included studies such as study designs, participants, tools and measurement, and modes of delivery prevents our efforts to conduct meta-analysis.

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INTRODUCTION

Coronavirus disease, or COVID-19, was first reported in December 2019(1). By March 2020, the World Health Organization (WHO) declared the COVID-19 a pandemic and as of November 2021, there have been over 259 million cumulative confirmed cases and 5 million deaths from COVID-19 worldwide(2). The rapid global spread of disease has not only had physical health impacts, but also psychological consequences on the population. Decreased population mobility and high rates of COVID-19 infection were found to be significantly related to an increased prevalence of major depression disorder. Compared to pre-pandemic, there were an additional 53.2 million cases of major depressive disorder and 76.2 million cases of anxiety disorders in 2020.

The COVID-19 outbreak has had direct and indirect impacts on mental health. Many experienced fear and anxiety due to uncertainty about the pandemic's evolution, feelings of hopelessness, despair, and grief in the face of uncontrollable events(3,4). The impact of public health measures such as quarantine measures, social restrictions, and isolation have also resulted in mental health consequences including anxiety, depression, and feelings of neglect; anxiety was highly prevalent in people undergoing quarantine or social isolation(5–7). The pandemic's indirect effects, such as unemployment due to economic downturn and the death of family members from disease, has led to depression and suicide(8–10).

Psychological consequences from the pandemic also developed more significantly in certain groups of people. Long hours of COVID-19 clinical services caused significant pandemic-related burnout for frontline health care workers worldwide(11–13). Fear of transmitting the infection to family members and increased demand for health services also caused stress and sleep problems among medical professionals(14–16).

Multiple mental health interventions have been implemented during previous epidemics such as Ebola and Middle East Respiratory Syndrome, particularly for health care workers experiencing higher levels of burnout and psychological consequences than others(17). Effective interventions like psychosocial and administrative support in workplaces have been shown to reduce the rates of

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emotional exhaustion in health care workers(17). Interventions in other population groups during Ebola outbreaks reduced depression, stress, and anxiety(18–20). Implementing interventions to alleviate mental health problems during the COVID-19 pandemic has proven to be difficult due to various social restrictions; face-to-face interventions seems not feasible where online interventions including telehealth and multimedia application were used as alternatives(21–23).

Though various interventions to mitigate psychosocial consequences from the COVID-19 pandemic have been launched, the tools used, their delivery channels and effectiveness on population groups are not well understood. This systematic review assesses interventions and tools used, and the effectiveness of interventions in mitigating psychological consequences on different populations so that effective interventions can be up-scaled accordingly.

We propose three review questions. First, what interventions are used to mitigate psychological consequences in the population and in health care workers? Second, what tools are used for assessing intervention outcome? Third, what is the effectiveness of different interventions? Findings from this review can inform health professionals and governments of effective interventions to prevent or mitigate mental health problems from the COVID-19 pandemic.

METHODS

Search strategy

We used PICO, a common tool used to frame research questions recommended by the Cochrane Collaboration(24). Population are individuals in countries or areas affected by COVID-19; interventions are exposure to COVID-19 pandemic and the clinical or non-clinical interventions; comparison of outcomes between case and control groups or before and after interventions; and outcome are mental consequences including suicide, depression, anxiety, stress, and insomnia/reduced sleep quality. See the full search strategy in Annex 1.

Published literatures were electronically retrieved from four databases including Medline, PubMed, Embase and Psycinfo. The eligibility criteria are English-language studies from 2020 onwards where

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interventions were provided with measurable outcomes. Table 1 provides search terms, inclusion and exclusion criteria.

Table 1 Search terms, inclusion and exclusion criteria

	Inclusion and exclusion criteria
Population	Inclusion: Individuals in countries or areas affected by COVID-19 including patients, health care workers, women, minority groups Exclusion: ● Individuals in countries or areas not affected by the COVID-19. ● Individuals having prior psychological symptoms
Intervention/ exposure	● Exposure to COVID-19 pandemic ● Both clinical and non-clinical interventions, provided by therapists, psychotherapists/counsellors, psychologists and qualified mental health staff including music and physical activity.
Comparison	● Comparing prevalence or degree of psychological consequences before and after interventions in the same group. ● Comparing prevalence or degree of psychological consequences in control and intervention groups.
Outcome	Mental health consequences including suicide, depression, anxiety, stress, and insomnia/reduced sleep quality. Outcomes must be quantified as proportions/prevalence rate or defined categorically.
Study design	Experimental research designs: ● Randomised controlled trials (RCTs)

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	• Controlled before and after studies (CBAs), with baseline and post-intervention measurement for both groups Exclusion • Type of publication: editorials, commentary, letters to the editor, reviews • Studies that did not have the outcomes measures specified in the protocol.

Selection of publications

Our search strategies identified a total of 5,623 records, of which 5,619 from the four databases and four from hand-searching from references of key articles which are in line with the research strategies and inclusion criteria.

After abstract screening, 705 duplicated records were removed while 4,602 records were not relevant. The full text of the remaining 316 records were retrieved and reviewed for eligibility. We found that 289 publications were not relevant to the review questions. Finally, 27 articles comprised of 29 studies were included for systematic review. Figure 1 shows the PRISMA flow of this process.

<Figure 1>

Figure 1 PRISMA Flow diagram of the review process

* Some studies measure multiple outcomes

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In the process of abstract screening, three researchers (AL, PP and AC) independently reviewed the abstracts. Consensus was reached if two researchers agreed based on eligibility criteria. In case of disagreement, the third researcher reviewed and decided.

Assessment the risk of bias

The Cochrane risk of bias tool (24) was applied to assess the methodological quality of 14 randomised controlled trials in this systematic review. There are six domains of risk of bias: 1) Random sequence generation (selection bias), 2) Allocation concealment (selection bias), 3) Blinding of participants, personnel and outcome assessors, 4) Incomplete outcome data (attrition bias), 5) Selective reporting (reporting bias) and 6) Other bias(24). Three authors (AS, PP and AC) separately evaluated each domain and categorised the risk of bias into 'low risk', 'high risk', or 'unclear risk'. Different criteria were used for making risk of bias judgements for each domain. Insufficient information reported in studies resulted in 'unclear risk' of bias judgment. When there were conflicting views, discussion among the three authors was convened to reach consensus.

Data extraction and synthesis

Relevant contents in all included articles were extracted and synthesised into three variables in an Excel spreadsheet: a) characteristic of study: author, year of publication, objective, country, study design, b) interventions: provider, participant, and c) outcomes: psychological consequence, and tool used for assessment of psychological consequence.

Patient and public involvement

No patients were involved in this study.

RESULTS

Study characteristics

Characteristics of 27 included articles are described in Table 2. One paper reported three studies, totalling 29 studies; all of which were conducted in 2020. Of the 29 studies, one was conducted globally(25) and the rest were categorised according to WHO geographical area. Three studies were conducted in the Eastern Mediterranean region(26–28), 10 in the European region(29–38), four in

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the Americas(39–42) and 11 in the Western Pacific(43–51). Applying the World Bank's income group categories, 13 studies were in high income(29–31,33–39,41,42,44), 12 in upper-middle income (32,40,43,45–51), and three in lower-middle income groups(26–28); there were no studies from the low income group.

On population, eight studies targeted COVID-19 patients(26–28,44,47,48,50,51), six among health care workers (29,32,33,42,45,46), and almost half of the studies were general population(15,25,30,31,34–41,43). Out of 29 studies, 14 were randomised control trials (RCT) (26–28,32,33,37–41,43,47,49). The remaining 15 were non-RCT studies.

The interventions reported by the studies can be classified into three types: 19 online-based interventions(25,27–29,31,33,34,36,38–44,46,47), six on-site-based interventions(26,32,37,48–50) and four combined online and on-site interventions(30,35,45,51).

On outcome measurement, only one study assessed suicide ideation(44), while 14 studies measured depression outcome(30,33,36–39,41,42,44–46,49–51), 24 studies measured anxiety(26,27,30,32–34,36–51), nine studies measured psychological stress(28,33–38,41,49), nine studies measured sleep quality(25,31,33,41,44,46–48,51), and six studies measured other outcomes such as loneliness and self-compassion(29,34,39,42,43). Table 2 shows different tools used to measure the outcomes in detail.

Table 2 Characteristics of the 29 included studies

Year of study	
- 2020	29/29*** (100%)
Geographic area (WHO regions*)	
- Eastern Mediterranean Region	3/29 (10.3%)
- European Region	10/29 (34.5%)
- Region of the Americas	4/29 (13.8%)
- Western Pacific Region	11/29 (37.9%)
- Global	1/29 (3.5%)
Income groups (World bank)**	
- High-income economies	13/29 (44.8%)
- Upper-middle-income economies	12/29 (41.4%)

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- Lower-middle-income economies	3/29 (10.3%)
- Global	1/29 (3.5%)
Participants	
- COVID-19 patients (confirmed/suspected)	8/29 (27.6%)
- Health care workers	6/29 (20.7%)
- General population (included student and teacher)	15/29 (51.7%)
Study design	
- Randomised control trial	14/29 (48.3%)
- Non-randomised control trial	15/29 (51.7%)
Intervention	
- Online	19/29 (65.5%)
- Onsite	6/29 (20.7%)
- Combined online and onsite	4/29 (13.8%)
Psychological outcome	
- Suicidal idea	1/29 (3.5%)
- Depression	14/29 (48.3%)
- Anxiety	24/29 (82.8%)
- Stress	9/29 (31.0%)
- Insomnia/ sleep quality	9/29 (31.0%)

* WHO regions: https://www.who.int/countries

** World bank data: https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups

*** There are 29 studies out of 27 papers

Tools used for the assessment of psychological consequences

A total of 20 tools were used to assess psychological outcomes in the 29 studies. We categorized the tools based on the outcome they measured including suicidal ideation, depression, anxiety, stress, and insomnia (Table 3, full description in Annex 2).

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175 Table 3 Tools for the assessment of psychological consequences

Outcomes	Tools	Method description	Studies
1. Suicidal Ideation Assessment	1.1 Beck Depression Inventory (BDI) (item 9)	A 21-item, self-reported rating inventory that measures attitudes and symptoms of depression. Item-9 of the test asked the potential to kill oneself.	Kim et al., 2020(44)
2. Depression Assessment	2.1 Patient Health Questionnaire-9 (PHQ-9)	A 9-item assessment that is used to assess the frequency the patients experience depressive symptoms according to DSM-IV depression criteria	Niu et al., 2021(50) Yang et al., 2020(51)
	2.2 Patient Health Questionnaire for Depression-8 (PHQ-8)	An 8-item assessment that consists of 8 of the 9 criteria on which the DSM-IV diagnosis of depressive disorders is based.	Kahlon et al., 2021(39)
	2.3 Patient Health Questionnaire-4 (PHQ-4)	A 4-item questionnaire that aims to allow for brief and accurate measurement of core symptoms or signs of depression and anxiety.	Sylvia et al., 2021(42)
	2.4 Depression Anxiety Stress Scale (DASS-21)	A 21-item test, short version of the original self-report questionnaire (DASS-42 item) to evaluate depression, anxiety, and stress symptoms with 7 items per subscale; DASS-21 Depression, (DASS-21 Anxiety, and DASS-21 Stress.	Riva et al., 2021(36), Wadhen et al., 2021(38), Fiol-DeRoque et al., 2021(33)
	2.5 Hospital Anxiety and Depression Scale (HADS)	A 14-item assessment used to measure anxiety and depression symptoms in medical patients. The HADS produces two scales with 7 items in each scale; one for anxiety (HADS–A) and one for depression (HADS–D).	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (HADS-D)(44)
	2.6 Self-rating Depression Scale (SDS)	A 20-item assessment to assess the subjective symptoms of depression for up to one week.	Liu et al., 2021(46), Li et al., 2020(45)
	2.7 Brief Symptom Inventory 18 (BSI-18)	An 18-item questionnaire, taken from the 53-item Brief Symptom Inventory (BSI). The scale includes three subscales (somatization, depression, and anxiety)	Zhang et al., 2021(49)
3. Anxiety assessment	3.1 Spielberger State-Trait Anxiety Inventory (STAI)	A 40-item self-report tool consisting of two parts for measuring state and trait anxiety with 20 questions in each section	Parizad et al., 2021(27), Khademi et al., 2021(26), González-García et al., 2021(34), Liu et al., 2020(47), Dincer et al., 2021(32), Guan et al., 2021(43)
	3.2 Anxiety - Generalized Anxiety Disorder scale (GAD-7)	A 7-item questionnaire to detect symptoms of generalized anxiety disorder based on seven core symptoms and assess the frequencies with which respondents experience these symptoms within the prior 2 weeks.	Kahlon et al., 2021(39), Luo et al., 2021(48), Niu et al., 2021(50), Yang et al., 2020(51)
	3.3 Self- Rating Anxiety Scale (SAS)	A 20-item assessment evaluating a variety of anxiety symptoms, including both psychological and somatic symptoms, that respondents experience within the past month.	Liu et al., 2021 (46), Guan et al., 2021(43), Li et al., 2020(45)
	3.4 Behaviour Assessment Scale for Children-3rd edition, (BASC III)	A test to identify and manage behavioral and emotional strengths and weaknesses of children and adolescents from three perspectives: teacher, parent and self (starting at age 8).	Malboeuf-Hurtubise et al., 2021)40(

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Outcomes	Tools	Method description	Studies
	3.5 Patient Health Questionnaire-4 (PHQ-4)	See 2.3	Sylvia et al., 2021 (42)
	3.6 Depression Anxiety Stress Scale (DASS-21)	See 2.4	Riva et al., 2021(36), Fiol-DeRoque et al., 2021(33), Wadhen et al., 2021(38)
	3.7 Hospital Anxiety and Depression Scale (HADS)	See 2.5	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (44)
	3.8 Brief Symptom Inventory 18 (BSI-18)	See 2.7	Zhang et al., 2021 (49)
4. Stress assessment	4.1 The General Health Questionnaire (GHQ-12)	A 12-item test, self-administered to evaluate mental distress and common mental disorders. There were multiple versions of GHQ including GHQ-60, GHQ-30, GHQ-28, and GHQ-12.	Gorbeña et al., 2021 (35)
	4.2 Perceived stress 14-item (PSS-14)	A 14-item scale used as a psychological instrument to measure the perception of stress, with seven positive items and seven negative items.	Wadhen et al., 2021(38), Shaygan et al., 2021 (28)
	4.3 Perceived Stress Scale 10-item (PSS-10)	A 10-item questionnaire about feelings and thoughts during the last month.	Riva et al., 2021(36), Solianik et al., 2021(37), Smith et al., 2021(41), González-García et al., 2021(34)
	4.4 Depression Anxiety Stress Scale (DASS-21)	See 2.4	Fiol-DeRoque et al., 2021(33)
	4.5 Brief Symptom Inventory 18 (BSI-18)	See 2.7	Zhang et al., 2021 (49)
5. Insomnia assessment	5.1 Insomnia Severity Index (ISI)	A 7-item assessment to assess current sleep characteristics.	Tunuguntla et al., 2021(25), Philip et al., 2020(31), Fiol-DeRoque et al., 2021(33), Kim, et al., 2020(44)
	5.2 St. Mary's Hospital Sleep Questionnaire (SMH)	A 14-item test asking about patient's sleep experience and comprises items to demonstrable reliability, with some open-ended questions.	Luo et al., 2021 (48)
	5.3 Patient-Reported Outcomes Measurement Information System (PROMIS) Sleep Disturbance Short Form	Items focus on perceptions of sleep quality, sleep depth, and restoration associated with sleep, perceived difficulties with getting to sleep or staying asleep, and perceptions of the adequacy of and satisfaction with sleep.	Smith et al., 2021 (41)
	5.4 Pittsburgh Sleep Quality Index (PSQI)	An 18-item index measuring seven sleep dimensions.	Liu et al., 2021(46), Yang et al., 2020(51)
	5.5 Sleep State Self-Rating Scale (SRSS)	A 10-item assessment used to assess sleep state.	Liu et al., 2020(47)

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Suicidal ideation was assessed by using one tool, the Beck Depression Inventory (BDI) item 9, whereas depression and anxiety were assessed by using seven and eight tools respectively. The Hospital Anxiety and Depression Scale (HADS) was most used for the assessment of depression, while the Spielberger State-Trait Anxiety Inventory (STAI) was used by six out of 24 studies on anxiety(26,27,32,34,43,47). Two different tools were used to assess both depression and anxiety, including the Patient Health Questionnaire-4 (PHQ-4), and HADS. The Brief Symptom Inventory-18 (BSI-18) and Depression Anxiety Stress Scale (DASS-21) were used to assess depression, anxiety, and psychological distress.

Stress was evaluated primarily using five tools, though the Perceived Stress Scale 10-item (PSS-10) was most commonly used. Among the other outcomes, insomnia assessments were distinctive in that they consisted of five tests, each of which measured the patients' sleep quality. The Insomnia Severity Index was widely used to determine sleep quality in four studies(25,31,33,44). There were several tools assessing outcomes beyond the focus of this study, one test for self-compassion and two tests for loneliness(34,39,42,43).

Interventions, measurements and effectiveness

Table 4 and 5 describes study design, providers of intervention, target population, intervention, measurement and psychological outcome (Full description in Annex 3).

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194 Table 4 Characteristics of the reviewed Randomised controlled trial (RCT) studies, intervention, measurement and psychological outcome

Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Dincer and Inangil, Turkey(32)	Certified emotional freedom technique	Nurses	Activating and deactivating signals to brain by stimulating points on skin	N = 72 Ni = 35 Nc = 37	Anxiety	STAI (+)
Fiol-DeRoque et al., Spain(33)	Self-conducted (online application)	Health care workers	PsyCovidApp, targeting emotional skills, healthy lifestyle behaviour, burnout, and social support	N = 436 Ni = 248 Nc = 234	Depression Anxiety Stress Insomnia	PHQ-8 (na) GAD7 (na) DASS-21 (na) ISI (na)
Guan, et al. (Study 2B), China(43)	Self-conducted (online)	General population	Self-compassion writing task	N = 79 Ni = 40 Nc = 39	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Guan et al. (Study 3), China(43)	Self-conducted (online)	College students	Self-compassion meditation and writing tasks	N = 95 Ni = 50 Nc = 45	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Kahlon et al., US(39)	Trained callers (Telephone-based)	Homebound older adults and people with disabilities	Phone call with empathetic conversations	N = 240 Nc = 120 Ni = 120	Depression Anxiety Loneliness	GAD7 (+) GAD7 (+) UCLA Loneliness Scale (+), De Jong Loneliness Scale (na)
Khademi et al., Iran(26)	Self-conducted	COVID-19 patients	Performed mandala colouring	N = 70 Ni = 35 Nc = 35	Anxiety	GAD-7 (+)
Liu et al., China(47)	Trainer	COVID-19 patients	Jacobson's relaxation techniques	Intervention = 51 Ni = 25 Nc = 26	Anxiety Insomnia	STAI (+) SRSS (+)

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Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Malboeuf-Hurtubise et al., Canada)40(	Trained students in psychology (Video)	School students	Mindfulness-based interventions and philosophical discussions on COVID-19	N = 37 (Pre-post intervention assessment)	Anxiety	BASC III (+)
Parizad et al., Iran(27)	Nurses	COVID-19 Patients	Guided imagery session with five audio tracks	N = 110 Nc = 55 Ni = 55	Anxiety	STAI (+)
Shaygan et al., Iran)28(	Self-conducted (online multimedia)	COVID-19 patients	Cognitive behavioural techniques, stress management, mindfulness, and positive psychotherapy	N = 48 Ni = 26 Nc = 22	Stress	PSS (+)
Smith et al., US)41(	Self-conducted (online application)	Obstetric and gynaecology patients	Mobile meditation app: encouragement, mindfulness meditation, sleep stories, nature sounds	N = 101 Ni = 50 Nc = 51	Depression Anxiety Stress Insomnia	HADS (+) HADS (+) PSS (+) PROMIS (+)
Solianik et al., Lithuania)37(	Tai chi instructor (online)	General population >60 years	Tai chi practice	N = 30 Ni = 15 Nc = 15	Depression Anxiety Stress	HADS (+) HADS (+) PSS-10 (+)
Wadhen et al., UK(38)	Yoga teacher (online)	General population	Yoga class via zoom meetings	N = 34 Nc = 17 Ni = 17	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (na), PSS (na)
Zhang et al., China(49)	Self-conducted (Online)	Residents	Mindfulness-Based Stress Reduction program	N = 51 Ni = 29 Nc = 28	Depression Anxiety Stress	BSI-18 (+) BSI-18 (+) BSI-18 (+)

195 + = Statistically significant association with post-intervention mental health improvements.

196 Table 5 Characteristics of the reviewed Non-randomised controlled study (NRS) studies, intervention, measurement and psychological outcome

Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Kim et al., South Korea(44)	Psychiatrists (Telephone-based)	COVID-19 patients	Ward telephone providing information and support for COVID-19 infection and hospital isolation	Week1 n=33 Week2 n=19 (Pre-post intervention assessment)	Suicidal idea Depression Anxiety Insomnia	BDI: week1 (+), week2 (na) HADS: week1 (+), week2 (na) HADS: week 1 (+), week 2 (na) ISI: week 1 (+), week 2 (na)
Guan, et al. (Study 2A), China(43)	Self-conducted (online)	Young adults	A live session of self-compassion meditation	N = 64 (Pre-post intervention assessment)	Anxiety	PHQ-4 (+)
Giordano et al., Italy(29)	Trained music therapist	Clinical staff	Music therapy supplemented by guided imagery and music	N = 34 (Pre-post ntervention assessment)	Others: fear/ worry	MTC (+)
González-García et al., Spain(34)	Mindfulness teacher and psychologist (online)	University students	Minilectures, mindfulness and compassion practices and self-reflection exercises	N = 66 (Pre-post intervention assessment)	Anxiety Stress Self-compassion	DASS-21 (+) DASS-21 (+) DASS-21 (+)
Gorbeña, et al., Spain(35)	Facilitators (interview and music playlists)	University students	Well-being and Personal Development Program	N = 151 Ni = 112 Nc= 39	Stress	PSS (+)
Li et al., China(45)	Teacher (Video)	Nurses	Simulation training about using personal protective devices	N = 60 (Pre-post intervention assessment)	Depression Anxiety	SDS (+) SAS (+)
Liu et al., China(46)	Self-conducted	Nurses	Diaphragmatic breathing relaxation training	N = 140 (Pre-post intervention assessment)	Depression Anxiety Insomnia	SDS (na) SAS (+) PSQI (+)
Luo et al., China(48)	Nurses	COVID-19 patients	Auricular point pressure	N = 84 Ni = 66	Anxiety	GAD-7 (+)

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Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
				Nc = 18	Insomnia	SMH score (+)
Matiz et al., Italy(30)	Socio-health educator and psychologist	School teachers	Mindfulness-oriented meditation training program	N = 67 (Pre-post intervention assessment)	Depression Anxiety	HADS (+) HADS (+)
Niu et al., China(50)	Nurses	(Suspected) COVID-19	Providing information about isolation, treatment, and social emotional support	N = 137 (Pre-post intervention assessment)	Depression Anxiety	PHQ-9 (+) GAD-7 (+)
Philip et al., France(31)	Self-conducted (online application)	General population	KANOPEE app with virtual agent interaction	N = 47 (Pre-post intervention assessment)	Insomnia	ISI (+)
Riva et al. Italy)36(	Self-conducted	General population	360◦ VR video entitled 'The Secret Garden' and a series of social exercises	N = 38 (Pre-post intervention assessment)	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (+), PSS (+)
Sylvia et al., US(42)	Psychologists, physicians, social workers, nurses (online)	Health care workers	Stress management and resilience training, relaxation response program	N = 102 (Pre-post intervention assessment)	Depression Anxiety Self-compassion	PHQ-4 (+) PHQ-4 (+) SCS (+)
Tunuguntla et al., Global level(25)	Self-conducted (online)	General population	Yoga and meditation with sound therapy	N = 820 (Pre-post intervention assessment)	Insomnia	ISI (+)
Yang et al., China(51)	Psychotherapist and nurse (in-person and online)	COVID-19 patients	Psychotherapy, empathy, muscle and breath relaxation, and cognitive behavioural therapy	N = 35 (Pre-post intervention assessment)	Depression Anxiety Insomnia	PHQ-9 (+) GAD-7 (+) PSQI (+)

197 + = Statistically significant association with post-intervention mental health improvements.

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One study provides suicidal prevention through 30-minute telephone consultation with psychiatrists. Compared to baseline, a significant reduction in suicidal ideation was found a one week ($p = 0.045$) but there were no statistical differences a two-weeks intervention ($p = 0.3996$)(44)

Fourteen studies provided different interventions to reduce depression, of which 11 (78.6%) significantly reduced depression(30,36–39,41,42,45,49–51). Of 14 studies, five of them were RCT; four related to mindfulness and meditation-based stress reduction(41,49), focusing on mind-body exercises including yoga(38) and tai chi(37)and one provided empathetic conversations through phone calls(39). Of total nine interventions against stress, seven had shown significant positive outcomes in stress reduction(28,34–37,41,49). Of these seven, four and three are RCT and non-RCT designs. One study shows no association of interventions and outcome(38), while one study showed positive and negative results due to timing of assessment(44).

Of 24 anxiety interventions from 18 publications, 21 (87.5%) significantly lowered anxiety in intervention groups; 11 were RCT(26,27,32,37,39–41,43,47,49) and 10 quasi-experimental studies(30,34,42–46,48,50,51). Among the 11 RCT studies, most interventions were mindfulness and meditation-based(37,40,41,43,49). Two studies applied specific techniques such as Emotional Freedom Techniques(32) and progressive muscle relaxation and deep breathing techniques(47).

Seven out of nine stress reduction studies (77.78%) reported significant stress reduction(34,35), in which five RCT studies used mindfulness meditation-based stress reduction interventions(28,36,37,41,49).

Of nine studies on insomnia, seven found that the interventions significantly reduced insomnia or improved sleep quality (77.8%)(25,31,41,46–48,51). Of these nine studies, only two were RCT, with one using a mobile meditation application(41), and the other using progressive muscle relaxation and deep breathing techniques(47). Three studies reported the effectiveness of interventions on self-compassion; participants reported significantly higher perception of self-compassion than those in the control group(34,42,43). One study using pre- and post- assessments show improvement of loneliness(39).

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Table 6 summarises psychological outcomes from interventions. Interventions used by 29 studies had effects on more than one psychological conditions. In this review, there were total 57 results. Out of 14 results on reduced depressions, 11 showed positive results(30,36–39,41,42,45,49–51); two did not show association between intervention and outcome(33,46), and one show mixed results of association and not association with intervention due to different phase of assessment(44). Out of 24 results on minimised anxiety, 20 showed positive results(26,27,30,32,34,37,39–43,45–51); three studies showed no association(33,36,38) , and one study showed mixed results due to difference in the timing of outcome assessment(44).

Seven out of nine results on reduced stress show positive results(28,34,35,37,41). Two results showed no association with intervention. Seven out of nine results on reduced insomnia or improve quality of sleep show positive results(25,41,46–48,51). One showed no association(47) and the other shows mixed results(44). Finally, outcome on reduced suicide ideation showed mixed results.

Table 6 Comparison of psychological outcome from interventions

			Interventions				
No	Author	RCT	Suicidal idea	Depression	Anxiety	Stress	Insomnia/ sleep quality
1.	Dincer and Inangil (32)	Y			+		
2.	Fiol-DeRoque et al. (33)	Y		NA	NA	NA	NA
3.	Guan et al. (study 2b)(43)	Y			+		
4.	Guan et al. (study 3)(43)	Y			+		
5.	Kahlon et al.(39)	Y		+	+		
6.	Khademi et al.(26)	Y			+		
7.	Liu et al.(47)	Y			+		+
8.	Malboeuf-Hurtubise et al.)40(	Y			+		
9.	Parizad et al.(27)	Y			+		
10.	Shaygan, et al.)28(	Y				+	
11.	Smith, et al.)41(	Y		+	+	+	+
12.	Solianik, et al.)37(	Y		+	+	+	
13.	Wadhen, et al.(38)	Y		+	NA	NA	

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14.	Zhang, et al.(49)	Y		+	+	+	
15.	González-García et al.(34)	N			+	+	
16.	Gorbeña et al.(35)	N				+	
17.	Guan et al. (study 2a)(43)	N			+		
18.	Kim et al.(44)	N	±	±	±		±
19.	Li et al.(45)	N		+	+		
20.	Liu et al.)46(	N		NA	+		+
21.	Luo et al.(48)	N			+		+
22.	Matiz et al.(30)	N		+	+		
23.	Niu et al.(50)	N		+	+		
24.	Philip et al.(31)	N					+
25.	Riva et al.)36(	N		+	NA	+	
26.	Sylvia, et al.(42)	N		+	+		
27.	Tunuguntla, et al.(25)	N					+
28.	Yang, et al.(51)	N		+	+		+
	Total studies	29					
	Total result	57	1	14	24	9	9
	Positive result	45	-	11	20	7	7
	±	4	1	1	1	0	1
	No association	8	-	2	3	2	1

Note:

+ denotes Statistically significant association with post-intervention mental health improvements.

NA denotes No association between measure and post-intervention symptoms.

± denotes mixed results of association and no association, due to difference in timing of outcome measurement

Grey shade denotes not applicable.

Risk of bias

The quality of reporting for all 14 RCT was assessed using the Cochrane risk of bias tool. Table 7 shows the risk of bias by six domains. All these RCT studies had no risk of attrition bias (incomplete outcome data) or reporting bias (selective reporting). Of all 14 studies, five had low risk of bias for all six dimensions reflecting high quality RCT. Most studies had unclear risk on allocation concealment,

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random sequence generation and blinding participants and personnel(26,37,40,43,47). Two studies reported high risk of bias on blinding of participants and personnel(38,49); as both studies were conducted through online platform and the group memberships were not blind.

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Table 7 Risk of bias assessments for all 14 RCT studies

	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Incomplete outcome data	Selective reporting	Other bias	Number of low risk out of six criteria
1. Dincer and Inangil (32)	Low	Low	Low	Low	Low	Low	6
2. Fiol-DeRoque et al.(33)	Low	Low	Low	Low	Low	Low	6
3. Guan et al. (study 2b)(43)	Unclear	Unclear	Unclear	Low	Low	Low	3
4. Guan et al. (study 3)(43)	Unclear	Unclear	Unclear	Low	Low	Low	3
5. Kahlon et al.(39)	Low	Low	Low	Low	Low	Low	6
6. Khademi et al.(26)	Low	Unclear	Unclear	Low	Low	Low	4
7. Liu et al.(47)	Unclear	Unclear	Unclear	Low	Low	Low	3
8. Malboeuf-Hurtubise et al.)40(	Unclear	Unclear	Low	Low	Low	Unclear	3
9. Parizad et al.(27)	Low	Low	Low	Low	Low	Low	6
10. Shaygan, et al.)28(	Low	Low	Low	Low	Low	Low	6
11. Smith, et al.)37,41(	Low	Low	Unclear	Low	Low	Low	5
12. Solianik, et al.)37(	Unclear	Unclear	Low	Low	Low	Low	4
13. Wadhen, et al.(38)	Low	Unclear	High	Low	Low	Low	4
14. Zhang, et al.(49)	Low	Unclear	High	Low	Low	Low	4

DISCUSSION

Study characteristics

Serious psychological impacts of COVID-19 on the population and health care workers calls for immediate mitigate responses as reflected by how all 29 studies in this review were carried out in 2020 in all geographical continents except Africa and Australia. The majority (45.2%) of studies were from high income countries, with 37.0% and 11.1% in upper-middle and lower-middle income countries. Size of pandemic surges and resulting government travel and social restrictions, cultural norms, social support and coping mechanisms are key factors influencing psychological consequences. Therefore, the varieties of interventions in this review can be useful for adaptive application in different country contexts, although there is no study from low-income countries. Psychological impacts are caused not only by direct consequences of the pandemic(52–54), but also by public health and social containment measures used by governments, notably movement

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restrictions and lockdowns(55). Over 50% of the psychological interventions reviewed in this study were delivered to COVID-19 patients, health care workers and general population via an online platform, as on-site intervention was not possible in the pandemic context. The psychological consequences from this pandemic are diverse, requiring a wide range of interventions and tools for measuring subsequent psychological outcomes, including suicidal ideation, depression, anxiety, stress, and insomnia.

Measurement tools for the psychological assessment

All of the tools were screening rather than diagnostic. Several tools were validated, where Cronbach's alpha values were greater than or equal to 0.70, indicating high reliability. However, the sensitivity and specificity of each tool were not reported in any articles.

In this review, only one study(44) assessed and applied an intervention for suicidal ideation. The Beck Depression Inventory (BDI) can be applied to a wide age range of 13- to 80-year olds, with internal consistency ranging from Cronbach's α of 0.73 to 0.92(56), making it a common tool for depression screening(57). In other studies, BDI sensitivity was ≥ 0.70 and specificity was ≥ 0.73 in non-clinical samples(57). Item-9 in BDI asked respondents to rate their thoughts to commit suicide on a scale of 0-3(58); there is a suicidal ideation if the score is 1 or higher(44).

The Hospital Anxiety and Depression Scale (HADS) was the most commonly used tool for assessment in all four depression interventions. HADS is used to measure both anxiety and depression, HADS-A for Anxiety and HADS-D for Depression, and is effective and widely used in both clinical and non-clinical populations, with reliable internal consistency for both anxiety (mean Cronbach's α = 0.83) and depression (mean Cronbach's α = 0.82)(59). The sensitivity and specificity for both HADS-A and HADS-D were approximately 0.80 and the concurrent validity between BDI and HADS-total score was 0.73(59).

Six of the 24 articles(26,27,32,34,43,47) used the Spielberger State-Trait Anxiety Inventory (STAI) for anxiety assessment. STAI can assess both state anxiety, which reflects transitory anxiety to current conditions, and trait anxiety, which refers to a general tendency to react anxiously to perceived

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threats from outside(60). This review found two studies that assessed both state and trait anxiety (27,47). Three studies assessed only state anxiety(26,32,43), and one study assessed only the trait anxiety(34). STAI seemed to be a highly adaptable tool with high internal consistency, Cronbach's α 0.92 and 0.90 for the state and trait anxiety respectively(61); Ercan et al. suggested a cut-off value of 41 for STAI (sensitivity=78.3 and specificity=71.2)(62).

The Perceived Stress Scale (PSS) was also used to assess psychological stress in six out of nine interventions to minimize stress; PSS-10 (10 items) was used in four(34,36,37,41) while PSS-14 was used in two studies for measurement(28,38). The PSS-10 is a shorter version of the PSS-14, consisting of six negative and four positive items of the PSS-14. However, the PSS-4, another truncated PSS version consisting of four items (two negative and two positive), was primarily designed for phone interviews(63). The Cronbach's α for the PSS 14-item, 10-item, and 4-item versions were 0.89, 0.91, and 0.82, respectively(64).

The Insomnia Severity Index (ISI) was most used to assess the severity of insomnia; four studies(25,31,33,44) used ISI to assess participants' sleep quality. Although the Pittsburgh Sleep Quality Index (PSQI) is reliable and valid, it was not designed specifically to assess insomnia. The ISI, on the other hand, was designed primarily to assess insomnia symptoms and consequences (e.g., distress), as well as to measure clinical significance of outcomes to insomnia treatments(65). It has high internal consistency coefficients between the community (Cronbach α = 0.90) and clinical cases (Cronbach α = 0.91)(65). A cut-off score of ten is the threshold (86.1% sensitivity and 87.7% specificity) for detecting insomnia cases in community samples(65).

Most of the tools were self-assessment questionnaires that study participants could complete using a simple scoring system. Some tests, however, required a specialist to interpret scores and ensure accurate results. BASC-III, for example, required a system to calculate the final score and interpret results since the test aimed to assess emotional disorders, personality constructs, and behavioral problems in children and adolescents aged 2 to 21 years old from 3 test sources including teachers, parents, and self (age >=8)(66). The St. Mary's Hospital Sleep Questionnaire (SMH) was another

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assessment with a complex scoring system employing both Likert scale and open-ended questions(67).

To assess each outcome, many articles used different versions of the same tool that differed in the number of questions and scoring system. PHQ-4, PHQ-8, and PHQ-9 versions of the Patient Health Questionnaire for Depression (PHQ) were used in four studies (39,42,50,51). The PHQ-9 incorporates DSM-IV depression criteria and other leading depression symptoms, whereas the PHQ-8 removes item 9 of PHQ 9(68) because it is difficult to provide appropriate coping services, including psychiatric evaluation, in some contexts(69). In one study(42), the PHQ-4 was used as a brief measurement of depression and anxiety(70); the two questions of depression (PHQ-2) were derived from the first two questions of PHQ-9, which asks the degree to which an individual has experienced depressive mood and anhedonia over the past two weeks(71) while the other two questions, known as the Generalized Anxiety Disorder Scale-2(GAD-2) was used for anxiety measurement. GAD-2 included the first two questions from the Generalized Anxiety Disorder Scale-7(GAD-7) that asks about the frequency of feeling nervous and being unable to stop or control worrying over the past two weeks(72). The PHQ-4 is a primary screening tool for depression; patients who screen positive should be further evaluated by PHQ-9 and GAD-7 to ensure an accurate result(71,72).

Interventions and psychological outcomes

Overall, most interventions have positive impact in mitigating psychological consequences in COVID-19 patients, health care workers and general population. A singular intervention can have combined effects for multiple psychological outcomes; for example one RCT using mobile meditation application including verbal and written encouragement, mindfulness meditation, sleep stories, nature sounds has significant positive effects on reduction of depression, anxiety, stress and insomnia(41). However, another study using an online application targeting emotional skills, healthy lifestyle behaviour, burnout, and social support reported no significant difference in the same measured outcomes(33).

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Almost all 14 RCT studies that included mindfulness and meditation interventions to reduce depression, anxiety and stress showed significantly improved outcomes (depression(37–39,41,49), anxiety(26,27,32,37–41,43,49), stress(37,41,49)) with the exception of one study (38), which did not improve anxiety. Stress, anxiety, and depression is caused by sympathetic nervous system activation due to widespread depolarization throughout the brain and body; on the other hand, meditation and deep breathing lead to parasympathetic nervous system activation due to widespread inhibition and hyperpolarization(73). Recently, mindfulness and meditation-based interventions have increasingly been integrated into mental health interventions(74); meditation practices facilitate training of the self-regulation of attention and awareness, thereby enhancing control of mental processes, and consequently increasing wellbeing. Meditation and breathing techniques such as yoga, tai-chi, targeting body and mind are also applied and are effective in reducing stress, anxiety, and depression(75,76). Yoga, for example, helps with intellectual and mental exercise, improves the psychological management and monitoring of stress and negative emotions, and supports mental balance(76–78). Mindfulness and meditation-based interventions are readily available, and do not pose threats or risks. It can be applied as a first-line or complementary intervention for stress, anxiety, depression, and some emotional disorders during COVID-19.

Moreover, combining mindfulness intervention with online application technology can benefit people who are quarantined or in isolation wards with no direct access to psychotherapy. In this review, online multimedia psychoeducational interventions were demonstrated to significantly reduce stress(28); mindfulness-based stress reduction, cognitive–behavioural techniques, stress management techniques, and positive psychotherapy have been delivered via WhatsApp. Nowadays, internet-based, self-administrated multimedia and education can promote access to mental health interventions, particularly amidst COVID-19. Online-based interventions enable people to social distance and be time-efficient and cost-effective.

The pandemic does not allow in-person on-site intervention, causing a major shift to remote care(79–81) and telemedicine; the use of electronic communication to deliver health care services at

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a distance, becomes a key supporting tool for health services delivery systems during this pandemic(82,83). Several channels of telemedicine are accessible by patients and health professionals including telephone calls, video calls, interactive mobile health applications (m-Health), short message service, e-mail, secure remote patient monitoring, and video conferencing. Video calls via social media is the most common channel of communication, but special communication softwares (e.g., Zoom) or current messenger applications (e.g., WhatsApp, and WeChat) are popular and convenient options as well(84).

Furthermore, several countries have developed digital health technology to facilitate health care delivery through an online platform(85,86). Several challenges remain; online interventions cannot replace in-person therapy, notably when there is high risk of suicide which necessitates the presence of a psychotherapist or human interaction(87,88). Furthermore, several online counsellors(89) raised concerns from their experiences, such as a lack of non-verbal language for interpreting patients who struggle to verbally communicate their feelings(90) or significantly shorter counseling periods through the telephone due to “more superficial responses to questions”(91). However, the pandemic has raised awareness for the necessity of the online interventions where limitations can be resolved, as several studies have reported benefits of online services(92–95).

Limitations

Causal interpretations are limited by 15 non-randomised control study design. In addition to variation in sample size across studies, there were also variations in the mode of delivery of interventions across different study sites. High heterogeneity of type of the study, participant and measurement tool limited the analysis. There was also a lack of assessment of co-interventions, which provide external sources of support to the participants in the intervention and control groups and may have influenced results of the outcomes of an intervention. There can be a large degree of reporting bias on psychological outcomes, especially when measuring outcome self-reporting and self-assessment. Further, there is no inclusion of cost-effectiveness assessment in any of the included all studies.

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CONCLUSION

Of the total 57 studied as psychological impacts of the pandemic, 45(78.9%) showed significant positive results--11 out of 14 for depression, 20 out of 24 for anxiety, 7 out of 9 for insomnia, 7 out of 9 for stress. Eight results showed no association with interventions and four show mixed results. Despite study limitations, the use of measurement tools and the strength of evidence from this review suggests an overall positive impact of the different interventions in terms of reducing psychological consequences in the population and health care workers. These effective interventions can be replicated and applied in other country settings, for which modes of delivery can be adjusted in line with country resources, epidemic situation, and social and health systems contexts. Moreover, it is necessary to improve the use of evidence in psychological interventions in mental health services.

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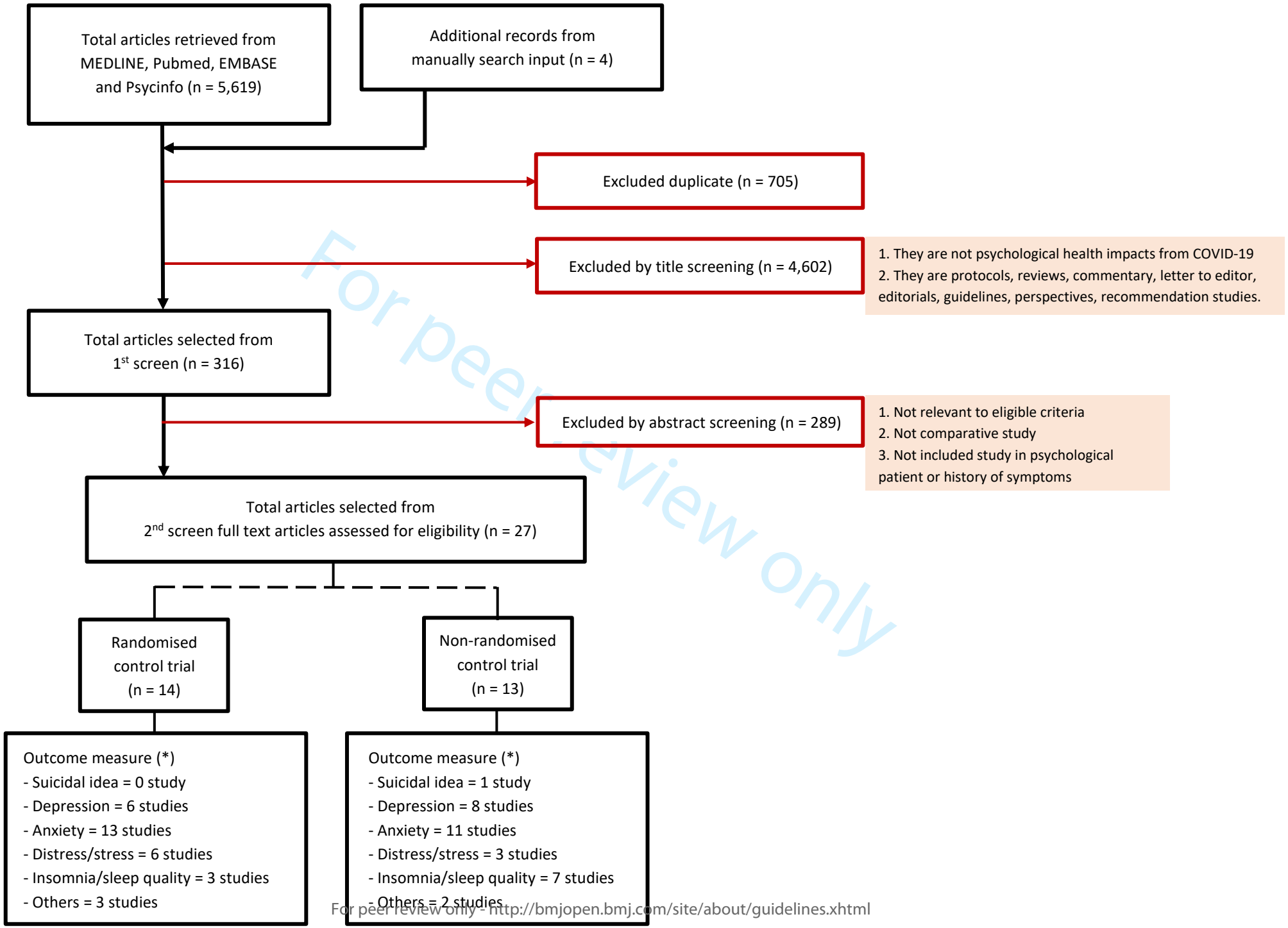
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Annex 1: Search term and results

1) PUBMED = 1,564
- COVID* OR novel corona* OR nCoV* OR 2019nCoV OR severe acute respiratory syndrome coronavirus 2 OR severe acute respiratory pneumonia outbreak OR SARS CoV 2 OR SARS-CoV-2 (206,274)
- therap*[MeSH Terms] OR intervention[MeSH Terms] OR consultation[MeSH Terms] (7,081,376)
- Mental Health [MeSH terms] OR Mental Disorders [MeSH terms] OR Mental Illness [MeSH terms] OR Suicide OR Depression OR Anxiety OR Psychological Distress [MeSH terms] OR Emotional Distress [MeSH terms] OR Insomnia [MeSH terms] (1,846,124)

2) EMBASE = 1,114
- COVID* (sh) or novel corona*.mp. or nCoV*.mp. or 2019nCoV.mp. or severe acute respiratory syndrome coronavirus 2 (sh) or severe acute respiratory pneumonia outbreak (sh) or SARS CoV 2.mp. or SARS-CoV-2.mp. (81,488)
- (treatment or therap*).mp. or intervention (sh) or consultation (sh) (13,205,742)
- (Mental Health or Mental Disorders or Mental Illness).mp. or Suicide (sh) or Depression (sh) or Anxiety (sh) or Psychological Distress.mp. or Emotional Distress.mp. or Insomnia (sh) (1,017,858)

3) PsycInfo = 1,524
- DE "Coronavirus" OR DE "COVID-19" OR DE "Severe Acute Respiratory Syndrome" (6,545)
- intervention or program or counseling (1,095,058)

4) MEDLINE(OVID) = 1,417
- (COVID* or novel corona* or nCoV* or 2019nCoV or severe acute respiratory syndrome coronavirus 2 or severe acute respiratory pneumonia outbreak or SARS CoV 2 or SARS-CoV-2).mp. (124,422)
- (therap* or therap* or intervention or consultation).mp. (6,269,680)
- (Mental Health or Mental Disorders or Mental Illness or Suicid* or Depression or Anxiety or Psychological Distress or Emotional Distress or Insomnia).mp. (638,321)

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Annex 2

Table A2 Tools for the assessment of psychological consequences with score classification

Outcomes	Tools	Method description	Score Classification	Studies
1. Suicidal Ideation Assessment	1.1 Beck Depression Inventory (BDI) (item 9)	A 21-item, self-reported rating inventory that measures attitudes and symptoms of depression. Item-9 of the test asked the potential to kill oneself (49).	Score 1-10: Normal Score 11-16: Mild Mood Disturbance Score 17-20: Borderline Clinical Depression Score 21-30: Moderate Depression Score 31-40: Severe Depression Score over 40: Extreme Depression	Kim et al., 2020(44)
2. Depression Assessment	2.1 Patient Health Questionnaire-9 (PHQ-9)	A 9-item assessment that is used to assess the frequency the patients experience depressive symptoms according to DSM-IV depression criteria(50).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score 15-19: Moderately Severe Score 20-27: Severe	Niu et al., 2021(50) Yang et al., 2020(51)
	2.2 Patient Health Questionnaire for Depression-8 (PHQ-8)	An 8-item assessment that consists of 8 of the 9 criteria on which the DSM-IV diagnosis of depressive disorders is based (51).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score 15-19: Moderately Severe Score 20-24: Severe	Kahlon et al., 2021(39)
	2.3 Patient Health Questionnaire-4 (PHQ-4)	A 4-item questionnaire that aims to allow for brief and accurate measurement of core symptoms or signs of depression and anxiety (52).	Score 0-2: Normal Score 3-5: Mild Score 6-8: Moderate Score 9-12: Severe	Sylvia et al., 2021(42)
	2.4 Depression Anxiety Stress Scale (DASS-21)	A 21-item test, short version of the original self-report questionnaire (DASS-42 item) to evaluate depression, anxiety, and stress symptoms with 7 items per subscale; (1) DASS-21 Depression, (2) DASS-21 Anxiety, and (3) DASS-21 Stress (53).	**Depression** Score 0-9: Normal Score 10-13: Mild Score 14-20: Moderate Score 21-27: Severe Score over 28: Extremely Severe **Anxiety** Score 0-7: Normal Score 8-9: Mild Score 10-14: Moderate Score 15-19: Severe	Riva et al., 2021(36), Wadhen et al., 2021(38), Fiol-DeRoque et al., 2021(33)

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Outcomes	Tools	Method description	Score Classification	Studies
			Score over 20: Extremely Severe **Stress** Score 0-14: Normal Score 15-18: Mild Score 19-25: Moderate Score 26-33: Severe Score over 34: Extremely Severe	
	2.5 Hospital Anxiety and Depression Scale (HADS)	A 14-item assessment used to measure anxiety and depression symptoms in medical patients. The HADS produces two scales with 7 items in each scale; one for anxiety (HADS–A) and one for depression (HADS–D) (54).	**Depression and Anxiety** Score 0-7: Normal Score 8-10: Borderline abnormal (borderline case) Score 11-21: Abnormal	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (HADS-D)(44)
	2.6 Self-rating Depression Scale (SDS)	A 20-item assessment to assess the subjective symptoms of depression for up to one week(55).	Score <50: Normal Score 50-59: Mild Depression Score 60-69: Moderate to Marked Major Depression Score >70: Extreme Major Depression	Liu et al., 2021(46), Li et al., 2020(45)
	2.7 Brief Symptom Inventory 18 (BSI-18)	An 18-item questionnaire, taken from the 53-item Brief Symptom Inventory (BSI). The scale includes three subscales (somatization, depression, and anxiety) (56).	The interpretation of the BSI-18 uses Global Severity Index (GSI) of distress represents sum across the 3 subscales range from 0-72. Higher scores indicate higher levels of psychological distress.	Zhang et al., 2021(49)
3. Anxiety assessment	3.1 Spielberger State-Trait Anxiety Inventory (STAI)	A 40-item self-report tool consisting of two parts for measuring state and trait anxiety with 20 questions in each section(57).	Score 20-31: Mild Score 32-42: Mild to Moderate Score 43-53: Moderate to Severe Score 54-64: Relatively Severe Score 65-75: Severe Score ≥ 76: Very Severe	Parizad et al., 2021(27), Khademi et al., 2021(26), González-García et al., 2021(34), Liu et al., 2020(47), Dincer et al., 2021(32), Guan et al., 2021(43)

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Outcomes	Tools	Method description	Score Classification	Studies
	3.2 Anxiety - Generalized Anxiety Disorder scale (GAD-7)	A 7-item questionnaire to detect symptoms of generalized anxiety disorder based on seven core symptoms and assess the frequencies with which respondents experience these symptoms within the prior 2 weeks (58).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score >=15: Severe	Kahlon et al., 2021(39), Luo et al., 2021(48), Niu et al., 2021(50), Yang et al., 2020(51)
	3.3 Self- Rating Anxiety Scale (SAS)	A 20-item assessment evaluating a variety of anxiety symptoms, including both psychological and somatic symptoms, that respondents experience within the past month (59).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score >=15: Severe	Liu et al., 2021 (46), Guan et al., 2021(43), Li et al., 2020(45)
	3.4 Behavior Assessment Scale for Children-3rd edition, (BASC III)	A test to identify and manage behavioral and emotional strengths and weaknesses of children and adolescents from three perspectives: teacher, parent and self (starting at age 8) (60).	Interpretation using the mean score. Higher scores indicate higher levels of anxiety.	Malboeuf-Hurtubise et al., 2021(40)
	3.5 Patient Health Questionnaire-4 (PHQ-4)	See 2.3		Sylvia et al., 2021 (42)
	3.6 Depression Anxiety Stress Scale (DASS-21)	See 2.4		Riva et al., 2021(36), Fiol-DeRoque et al., 2021(33), Wadhen et al., 2021(38)
	3.7 Hospital Anxiety and Depression Scale (HADS)	See 2.5	Score 20-44: Normal Score 45-59: Mild to Moderate Anxiety Score 60-74: Marked to Severe Anxiety levels	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (44)
	3.8 Brief Symptom Inventory 18 (BSI-18)	See 2.7		Zhang et al., 2021 (49)

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Outcomes	Tools	Method description	Score Classification	Studies
4. Psychological distress and stress assessment	5. The General Health Questionnaire (GHQ-12)	A 12-item test, self-administered to evaluate mental distress and common mental disorders. There were multiple versions of GHQ including GHQ-60, GHQ-30, GHQ-28, and GHQ-12 (61).	The average sum of its items provides a scalar indicator of the degree of mental distress. Score of 11 or 12: typical Score >15: Suggesting evidence of distress Score >20: Severe problems with psychological distress	Gorbeña et al., 2021 (35)
	4.1 Perceived stress 14-item (PSS-14)	A 14-item scale used as a psychological instrument to measure the perception of stress, with seven positive items and seven negative items (62).	Score 0-18: Low Stress Score 19-37: Moderate Stress Score 38-56: High Stress	Wadhen et al., 2021(38), Shaygan et al., 2021 (28)
	4.2 Perceived Stress Scale 10-item (PSS-10)	A 10-item questionnaire about feelings and thoughts during the last month(63) .	Score 0-13: Low Stress Score 14-26: Moderate Stress Score 27-40: High Perceived Stress	Riva et al., 2021(36), Solianik et al., 2021(37), Smith et al., 2021(41), González-García et al., 2021(34)
	4.3 Depression Anxiety Stress Scale (DASS-21)	See 2.4	Score 0-13: Low Stress Score 14-26: Moderate Stress Score 27-40: High Perceived Stress	Fiol-DeRoque et al., 2021(33)
	4.4 Brief Symptom Inventory 18 (BSI-18)	See 2.7		Zhang et al., 2021 (49)
5. Insomnia assessment	5.1 Insomnia Severity Index (ISI)	A 7-item assessment to assess current sleep characteristics(64).	Score 0-7: No clinically significant insomnia Score 8-14: Subthreshold Insomnia Score 15-21: Clinically Significant Insomnia (Moderate) Score 22-28: clinically Significant Insomnia (Severe)	Tunuguntla et al., 2021(25), Philip et al., 2020(31), Fiol-DeRoque et al., 2021(33), Kim, et al., 2020(44)
	5.2 St. Mary's Hospital Sleep Questionnaire (SMH)	A 14-item test asking about patient's sleep experience and comprises items to demonstrable reliability, with some open-ended questions(65).	The test needs to be evaluated by experts.	Luo et al., 2021 (48)

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Outcomes	Tools	Method description	Score Classification	Studies
	5.3 PROMIS (Patient-Reported Outcomes Measurement Information System) Sleep Disturbance Short Form	Items focus on perceptions of sleep quality, sleep depth, and restoration associated with sleep, perceived difficulties with getting to sleep or staying asleep, and perceptions of the adequacy of and satisfaction with sleep (66).	Score <55: None to slight Score 55-59.9: Mild Score 60-69.9: Moderate Score >70: Severe	Smith et al., 2021 (41)
	5.4 Pittsburgh Sleep Quality Index (PSQI)	An 18-item index measuring seven sleep dimensions (67).	The overall PSQI score ranges from 0 to 21, with higher scores reflecting poorer sleep quality. A revised PSQI score of 7 or higher is considered poor sleep quality. If the score of any individual dimension is higher than 1, there is a problem with that specific dimension.	Liu et al., 2021(46), Yang et al., 2020(51)
	5.5 Sleep State Self-Rating Scale (SRSS)	A 10-item assessment used to assess sleep state.	SRSS over or equal to 23 points is considered to have sleep problems.	Liu et al., 2020(47)

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Annex 3

Table A3 Characteristics of the reviewed studies, intervention, measurement and psychological outcome

Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Dincer and Inangil, Turkey(32)	RCT	Certified emotional freedom technique	Nurses	Activating and deactivating signals to brain by stimulating points on skin	N = 72 Ni = 35 Nc = 37	Anxiety	STAI (+)
Fiol-DeRoque et al., Spain(33)	RCT	Self-conducted (online application)	Health care workers	PsyCovidApp, targeting emotional skills, healthy lifestyle behaviour, burnout, and social support	N = 436 Ni = 248 Nc = 234	Depression Anxiety Stress Insomnia	PHQ-8 (na) GAD7 (na) DASS-21 (na) ISI (na)
Guan, et al. (Study 2B), China(43)	RCT		General population	Self-compas sion writing task	N = 79 Ni = 40 Nc = 39	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Guan et al. (Study 3), China(43)	RCT		College students	Self-compassion meditation and writing tasks	N = 95 Ni = 50 Nc = 45	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Kahlon et al., US(39)	RCT	Trained callers (Telephone-based)	Homebound older adults and people with disabilities	Phone call with empathetic conversations	N = 240 Nc = 120 Ni = 120	Depression Anxiety Loneliness	GAD7 (+) GAD7 (+) UCLA Loneliness Scale (+), De Jong Loneliness Scale (na)
Khademi et al., Iran(26)	RCT	Self-conducted	COVID-19 patients	Performed mandala colouring	N = 70 Ni = 35 Nc = 35	Anxiety	GAD-7 (+)
Liu et al., China(47)	RCT	Trainer	COVID-19 patients	Jacobson's relaxation techniques	Intervention = 51 Ni = 25	Anxiety Insomnia	STAI (+) SRSS (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					Nc = 26		
Malboeuf-Hurtubise et al., Canada(40)	RCT	Trained students in psychology (Video)	School students	Mindfulness-based interventions and philosophical discussions on COVID-19	N = 37 (Pre-post intervention assessment)	Anxiety	BASC III (+)
Parizad et al., Iran(27)	RCT	Nurses	COVID-19 Patients	Guided imagery session with five audio tracks	N = 110 Nc = 55 Ni = 55	Anxiety	STAI (+)
Shaygan et al., Iran(28)	RCT	Self-conducted (online multimedia)	COVID-19 patients	Cognitive behavioural techniques, stress management, mindfulness, and positive psychotherapy	N = 48 Ni = 26 Nc = 22	Stress	PSS (+)
Smith et al., US(41)	RCT	Self-conducted (online application)	Obstetric and gynaecology patients	Mobile meditation app: encouragement, mindfulness meditation, sleep stories, nature sounds	N = 101 Ni = 50 Nc = 51	Depression Anxiety Stress Insomnia	HADS (+) HADS (+) PSS (+) PROMIS (+)
Solianik et al., Lithuania(37)	RCT	Tai chi instructor (online)	General population >60 years	Tai chi practice	N = 30 Ni = 15 Nc = 15	Depression Anxiety Stress	HADS (+) HADS (+) PSS-10 (+)
Wadhen et al., UK(38)	RCT	Yoga teacher (online)	General population	Yoga class via zoom meetings	N = 34 Nc = 17 Ni = 17	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (na), PSS (na)
Zhang et al., China(49)	RCT	Self-conducted (Online)	Residents	Mindfulness-Based Stress Reduction program	N = 51 Ni = 29	Depression Anxiety	BSI-18 (+) BSI-18 (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					Nc = 28	Stress	BSI-18 (+)
Kim et al., South Korea(44)	NRS	Psychiatrists (Telephone-based)	COVID-19 patients	Ward telephone providing information and support for COVID-19 infection and hospital isolation	Week1 n=33 Week2 n=19 (Pre-post intervention assessment)	Suicidal idea Depression Anxiety Insomnia	BDI: week1 (+), week2 (na) HADS: week1 (+), week2 (na HADS: week 1 (+), week 2 (na) ISI: week 1 (+), week 2 (na)
Guan, et al. (Study 2A), China(43)	NRS	Self-conducted (online)	Young adults	A live session of self-compassion meditation	N = 64 (Pre-post intervention assessment)	Anxiety	PHQ-4 (+)
Giordano et al., Italy(29)	NRS	Trained music therapist	Clinical staff	Music therapy supplemented by guided imagery and music	N = 34 (Pre-post intervention assessment)	Others: fear/ worry	MTC (+)
González-García et al., Spain(34)	NRS	Mindfulness teacher and psychologist (online)	University students	Minilectures, mindfulness and compassion practices and self-reflection exercises	N = 66 (Pre-post intervention assessment)	Anxiety Stress Self-compassion	DASS-21 (+) DASS-21 (+) DASS-21 (+)
Gorbeña, et al., Spain(35)	NRS	Facilitators (interview and music playlists)	University students	Well-being and Personal Development Program	N = 151 Ni = 112 Nc= 39	Stress	PSS (+)
Li et al., China(45)	NRS	Teacher (Video)	Nurses	Simulation training about using personal protective devices	N = 60 (Pre-post intervention assessment)	Depression Anxiety	SDS (+) SAS (+)
Liu et al., China(46)	NRS	Self-conducted	Nurses	Diaphragmatic breathing relaxation training	N = 140 (Pre-post	Depression	SDS (na)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					intervention assessment)	Anxiety Insomnia	SAS (+) PSQI (+)
Luo et al., China(48)	NRS	Nurses	COVID-19 patients	Auricular point pressure	N = 84 Ni = 66 Nc = 18	Anxiety Insomnia	GAD-7 (+) SMH score (+)
Matiz et al., Italy(30)	NRS	Socio-health educator and psychologist	School teachers	Mindfulness-oriented meditation training program	N = 67 (Pre-post intervention assessment)	Depression Anxiety	HADS (+) HADS (+)
Niu et al., China(50)	NRS	Nurses	(Suspected) COVID-19	Providing information about isolation, treatment, and social emotional support	N = 137 (Pre-post intervention assessment)	Depression Anxiety	PHQ-9 (+) GAD-7 (+)
Philip et al., France(31)	NRS	Self-conducted (online application)	General population	KANOPEE app with virtual agent interaction	N = 47 (Pre-post intervention assessment)	Insomnia	ISI (+)
Riva et al. Italy(36)	NRS	Self-conducted	General population	360◦ VR video entitled 'The Secret Garden' and a series of social exercises	N = 38 (Pre-post intervention assessment)	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (+), PSS (+)
Sylvia et al., US(42)	NRS	Psychologists, physicians, social workers, nurses (online)	Health care workers	Stress management and resilience training, relaxation response program	N = 102 (Pre-post intervention assessment)	Depression Anxiety Self-compassion	PHQ-4 (+) PHQ-4 (+) SCS (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Tunuguntla et al., Global level(25)	NRS	Self-conducted (online)	General population	Yoga and meditation with sound therapy	N = 820 (Pre-post intervention assessment)	Insomnia	ISI (+)
Yang et al., China(51)	NRS	Psychotherapist and nurse (in-person and online)	COVID-19 patients	Psychotherapy, empathy, muscle and breath relaxation, and cognitive behavioural therapy	N = 35 (Pre-post intervention assessment)	Depression Anxiety Insomnia	PHQ-9 (+) GAD-7 (+) PSQI (+)

RCT: Randomised controlled trial

NRS: Non-randomised controlled study

+ = Statistically significant association with post-intervention mental health improvements.

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4, Line 73-81
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5, Line 87-90
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Table 1
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P 5, Line 100
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Figure 1
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P 5, Line 106
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Table 1
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P9, Line 135
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P9, Line 124
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Table 1
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	NA
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	P9, Line 124

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P9, Line 141
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	P9, Line 141, Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 4 and 5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table 4
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Table 6
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P27, Line 329
	23b	Discuss any limitations of the evidence included in the review.	P27, Line 380
	23c	Discuss any limitations of the review processes used.	P27, Line 380
	23d	Discuss implications of the results for practice, policy, and future research.	P27, Line 390
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Not registered
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P30, Line 402

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Competing interests	26	Declare any competing interests of review authors.	NA
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71
For more information, visit: http://www.prisma-statement.org/

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Living through the psychological consequences of COVID-19 pandemic: a systematic review of effective mitigating interventions

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Title: Living through the psychological consequences of COVID-19 pandemic: a systematic review of effective mitigating interventions

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Structured abstract 268 words
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Table 5
Figure 1
Reference 84
Key words: psychosocial consequences, COVID-19, pandemic

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ABSTRACT

Objective

This review assesses interventions and their effectiveness in mitigating psychological consequences from pandemic.

Method

Published English literature were searched from four databases [Medline, PubMed, Embase, and Psycinfo] from January 2020 and September 2021. Total 27 papers with 29 studies (one paper reported three studies) met inclusion criteria. Cochrane risk of bias tool is applied to assess quality of all randomized control trials (RCT).

Results

All studies were recently conducted in 2020. Publications were from by high- (13), upper-middle (12) and lower-middle income countries (3) and one global.Half of the studies conducted for general population (51.7%). One-third of studies (27.6%) provided interventions to COVID-19 patients and 20.7% to health care workers. Of 29 studies, 48.3% were RCT. All RCT were assessed for risk of biases; five studies (35.7%) had low risk as measured against all six dimensions reflecting high quality study.

Of these 29 studies, 26 diagnostic or screening measures were applied; eight for anxiety, seven for depression, five for stress, five for insomnia and one for suicide. Measures used to assess the baseline and outcomes of interventions were standardized and widely applied by other studies with high level of reliability and validity. Of 11 RCT studies, 24 anxiety interventions significantly lowered anxiety in intervention groups. Five RCT studies provided significantly reduced depression. Most interventions for anxiety and stress were mindfulness and meditation-based.

Conclusions

A majority, 78.9% of results from 29 interventions were effective in mitigating psychological consequences from COVID-19 pandemic when applied to health care workers, COVID-19 patients and general population. These effective interventions can be applied in other country settings through adaptation of modes of delivery suitable to country resources, pandemic and health systems context.

Article summary

Strengths and limitations of this study

- Fourteen randomized controlled trials provide robust asessment of the effectiveness of interventions.
- Interventions cover a variety of targets including patients, healhtcare workers and general population.
- The heterogeneity of included studies such as designs, measures and measurement, and modes of delivery prevents efforts to conduct meta-analysis.

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INTRODUCTION

Coronavirus disease, or COVID-19, was first reported in December 2019(1). By March 2020, the World Health Organization (WHO) declared the COVID-19 a pandemic and as of November 2021, there have been over 259 million cumulative confirmed cases and 5 million deaths from COVID-19 worldwide(2). The rapid global spread of disease has not only had physical health impacts, but also psychological consequences on the population.

The COVID-19 outbreak has had direct and indirect impacts on mental health. Many experienced fear and anxiety due to uncertainty about the pandemic's evolution, feelings of hopelessness, despair, and grief in the face of uncontrollable events(3,4). The impact of public health measures such as quarantine measures, social restrictions, and isolation have also resulted in mental health consequences including anxiety, depression, and feelings of neglect; anxiety was highly prevalent in people undergoing quarantine or social isolation(5–7). The pandemic's indirect effects, such as unemployment due to economic downturn and the death of family members from disease, has led to depression and suicide(8–10). Decreased population mobility and high rates of COVID-19 infection were found to be significantly related to an increased prevalence of major depression disorder, which compared to pre-pandemic, there were an additional 53.2 million cases of major depressive disorder and 76.2 million cases of anxiety disorders in 2020 (11).

Psychological consequences from the pandemic also developed more significantly in certain groups of people. Long hours of COVID-19 clinical services caused significant pandemic-related burnout for frontline health care workers worldwide(12–14). Fear of transmitting the infection to family members and increased demand for health services also caused stress and sleep problems among medical professionals(15–17).

Multiple mental health interventions have been implemented during previous epidemics such as Ebola and Middle East Respiratory Syndrome, particularly for health care workers experiencing higher levels of burnout and psychological consequences than others(18). Effective interventions like psychosocial and administrative support in workplaces have been shown to reduce the rates of emotional exhaustion in health care workers(18). Interventions in other population groups during Ebola outbreaks reduced depression, stress, and anxiety(19–21). Implementing interventions to alleviate mental health problems during the COVID-19 pandemic has proven to be difficult due to various social restrictions; face-to-face interventions seems not feasible where online interventions including telehealth and multimedia application were used as alternatives(22–24).

Though various interventions to mitigate psychosocial consequences from the COVID-19 pandemic have been launched, the measures used, their delivery channels and effectiveness on population groups are not well understood. This systematic review assesses interventions and measures used, and the effectiveness of interventions in mitigating psychological consequences on different populations so that effective interventions can be up-scaled accordingly. The psychological consequences affect patients who suffered from the illnesses and stress from quarantine and isolation, healthcare workers suffered from stressful long hour work and general population facing indirect effects of government measures such as lock down and limited social activities.

We propose three review questions. First, what interventions are used to mitigate psychological consequences in the population and in health care workers? Second, what measures are used for assessing intervention outcome? Third, what is the effectiveness of different interventions? Findings from this review can inform health professionals and governments of effective interventions to prevent or mitigate mental health problems from the COVID-19 pandemic.

METHODS

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Search strategy

We used PICO framework, a common tool used to to develop literature search strategies to ensure the comprehensive searches recommended by the Cochrane Collaboration(25). It is used in evidence-based practice to formulate healthcare related question. PICO elements include Population, Interventions, Comparisons and Outcomes. In this review, opulation are individuals in countries or areas affected by COVID-19; interventions are exposure to COVID-19 pandemic and the clinical or non-clinical interventions which mitigate the psychological impact from the pandemic; comparison of outcomes between case and control groups or before and after interventions; and outcome are mental consequences including suicide, depression, anxiety, stress, and insomnia/reduced sleep quality. See the full search strategy in Annex 1.

Published literatures were electronically retrieved from four databases including Medline, PubMed, Embase and Psycinfo. The eligibility criteria are English-language studies between January 2020 – September 2021where interventions were provided with measurable outcomes. Table 1 provides inclusion and exclusion criteria.

Table 1 Search terms, inclusion and exclusion criteria

	Inclusion and exclusion criteria
Population	Inclusion: Individuals in countries or areas affected by COVID-19 including patients, health care workers, women, minority groups Exclusion: ● Individuals in countries or areas not affected by the COVID-19. ● Individuals having prior psychological symptoms
Intervention/ exposure	Inclusion: ● Exposure to COVID-19 pandemic ● Both clinical and non-clinical interventions, provided by therapists, psychotherapists/counsellors, psychologists and qualified mental health staff including music and physical activity. Exclusion: Not any
Comparison	Inclusion: ● Comparing prevalence or degree of psychological consequences before and after interventions in the same group. ● Comparing prevalence or degree of psychological consequences in control and intervention groups. Exclusion: Not any
Outcome	Inclusion: Mental health consequences including suicide, depression, anxiety, stress, and insomnia/reduced sleep quality. Outcomes must be quantified as proportions/prevalence rate or defined categorically. Exclusion: Not any
Study design	Inclusion: Experimental research designs; ● Randomised controlled trials (RCTs) ● Controlled before and after studies (CBAs), with baseline and post-intervention measurement for both groups Exclusion:

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	• Type of publication: editorials, commentary, letters to the editor, reviews • Studies that did not have the outcomes measures specified in the protocol.

Selection of publications

The search terms were identified based on search strategy and applied to search in title, abstract, keyword and full-text. After searching, the duplicated studies or data were removed then the abstracts were reviewed; those were excluded for the following reasons with agreement in inclusion and exclusion criteria.

Figure 1 shows the PRISMA flow of this process.

<Figure 1>

Figure 1 PRISMA Flow diagram of the review process
Note * Some studies measure multiple outcomes

In the process of abstract screening, three researchers (AL, PP and AC) independently reviewed the abstracts. Consensus was reached if two researchers agreed based on eligibility criteria. In case of disagreement, the third researcher reviewed and decided.

Assessment the risk of bias

The Cochrane risk of bias tool (25) was applied to assess the methodological quality of 14 randomised controlled trials included by this systematic review. There are six domains of risk of bias: 1) Random sequence generation (selection bias), 2) Allocation concealment (selection bias), 3) Blinding of participants, personnel and outcome assessors, 4) Incomplete outcome data (attrition bias), 5) Selective reporting (reporting bias) and 6) Other bias(25). Three authors (AS, PP and AC) separately evaluated each domain and categorised the risk of bias into 'low risk', 'high risk', or 'unclear risk'. The results of assessment were showed in terms of number of 'low risk' of bias, which is six in total. Different criteria were used for making risk of bias judgements for each domain. Insufficient information reported in studies resulted in 'unclear risk' of bias judgment. When there were conflicting views, discussion among the three authors was convened to reach consensus.

Data extraction and synthesis

Relevant contents in all included articles were extracted and synthesised into three variables in an Excel spreadsheet: a) characteristic of study: author, year of publication, objective, country, study design, b) interventions: provider, participant, and c) outcomes: different types of psychological consequence, and measures used for assessment of psychological consequence.

Patient and public involvement

No patients were involved in this study.

RESULTS

Our search strategies identified a total of 5,623 records, of which 5,619 from the four databases and four from hand-searching from references of key articles. After abstract screening, 705 duplicated records were removed while 4,602 records were not relevant. The full text of the remaining 316 records were retrieved and reviewed for eligibility. We found that 289 publications were not relevant to the review questions. Finally, 27 articles comprised of 29 studies were included for systematic review.

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Study characteristics

Characteristics of 27 included articles are described in Table 2. One paper reported three studies (44), totalling 29 studies; all of which were conducted in 2020. Of the 29 studies, one was conducted globally (3.5%) (26) and the rest were categorised according to WHO geographical area. Three studies were conducted in the Eastern Mediterranean region (10.3%) (27–29), 10 in the European region (34.5%) (30–39), four in the Americas (13.8%) (40–43) and 11 in the Western Pacific (37.9%)(44–52). Applying the World Bank's income group categories, 13 studies were in high income (44.8%) (30–32,34–40,42,43,45), 12 in upper-middle income (41.4%) (33,41,44,46–52), and three in lower-middle income groups (10.3%)(27–29); there were no studies from the low income group.

On population, eight studies (27.6%) targeted COVID-19 patients (20.7%) (27–29,45,48,49,51,52), six among health care workers (30,33,34,43,46,47), and almost half of the studies were general population (51.7%) (,26,31,32,35–42,44,50). Out of 29 studies, 14 (48.3%) were randomised control trials (RCT) (27–29,33,34,38–42,44(2b,3),48,50). The remaining 15 were non-RCT studies.

The interventions reported by the studies can be classified into three types: 19 online-based interventions (65.5%) (26,28–30,32,34,35,37,39–45,47,48), six on-site-based interventions (20.7%) (27,33,38,49–51) and four combined online and on-site interventions (13.8%) (31,36,46,52).

On outcome measurement, only one study assessed suicide ideation (3.5%) (45), while 14 studies measured depression outcome (48.3%) (31,34,37–40,42,43,45–47,50–52), 24 studies measured anxiety (82.8%) (27,28,31,33–35,37–52), nine studies measured psychological stress(29,34–39,42,50), nine studies measured sleep quality (31.0%) (26,32,34,42,45,47–49,52), and six studies measured other outcomes such as loneliness and self-compassion (20.7%) (30,35,40,43,44). Table 2 shows characteristics of the studies.

Table 2 Characteristics of the 29 included studies

Geographic area (WHO regions)		
- Eastern Mediterranean Region	3 (10.3%)	27-29
- European Region	10 (34.5%)	30-39
- Region of the Americas	4 (13.8%)	40-43
- Western Pacific Region	11 (37.9%)	44-52
- Global	1 (3.5%)	26
Income groups (World bank)		
- High-income economies	13 (44.8%)	30-32, 34-40, 42, 43, 45
- Upper-middle-income economies	12 (41.4%)	33, 41, 44, 46-52
- Lower-middle-income economies	3 (10.3%)	27-29
- Global	1 (3.5%)	26
Participants		
- COVID-19 patients (confirmed/suspected)	8 (27.6%)	27-29, 45, 48, 49, 51, 52
- Health care workers	6 (20.7%)	30, 33, 34, 43, 46, 47
- General population (included student and teacher)	15 (51.7%)	26, 31, 32, 35-42, 44, 50
Study design		
- Randomised control trial	14 (48.3%)	27-29, 33, 34, 38-42, 44(2B,3), 48, 50

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- Non-randomised control trial	15 (51.7%)	26, 30-32, 35-37, 43, 44(2A), 45-47, 49, 51, 52
Intervention		
- Online	19 (65.5%)	26, 28-30, 32, 34, 35, 37, 39–45, 47, 48
- Onsite	6 (20.7%)	27, 33, 38, 49–51
- Combined online and onsite	4 (13.8%)	31, 36, 46, 52
Psychological outcome		
- Suicidal idea	1 (3.5%)	45
- Depression	14 (48.3%)	31, 34, 37–40, 42, 43, 45–47, 50–52
- Anxiety	24 (82.8%)	27, 28, 31, 33–35, 37–52
- Stress	9 (31.0%)	29, 34–39, 42, 50
- Insomnia/ sleep quality	9 (31.0%)	26, 32, 34, 42, 45, 47–49, 52

*** There are 29 studies out of 27 papers

Measures used for the assessment of psychological consequences

A total of 20 measures were used to assess psychological outcomes in the 29 studies. We categorized the measures based on the outcome they measured including suicidal ideation depression, anxiety, stress, and insomnia (Annex 2).

Suicidal ideation was assessed by using one measure, the Beck Depression Inventory (BDI) item 9, whereas depression and anxiety were assessed by using seven and eight measures respectively. The Hospital Anxiety and Depression Scale (HADS) was most used for the assessment of depression, while the Spielberger State-Trait Anxiety Inventory (STAI) was used by six out of 24 studies on anxiety(27,28,33,35,44,48). Two different measures were used to assess both depression and anxiety, including the Patient Health Questionnaire-4 (PHQ-4), and HADS. The Brief Symptom Inventory-18 (BSI-18) and Depression Anxiety Stress Scale (DASS-21) were used to assess depression, anxiety, and psychological distress.

Stress was evaluated primarily using five measures, though the Perceived Stress Scale 10-item (PSS-10) was most commonly used. Among the other outcomes, insomnia assessments were distinctive in that they consisted of five tests, each of which measured the patients' sleep quality. The Insomnia Severity Index was widely used to determine sleep quality in four studies(26,32,34,45). There were several measures assessing outcomes beyond the focus of this study, one test for self-compassion and two tests for loneliness(35,40,43,44).

Interventions, measurements and effectiveness

Table 3 and 4 describes study design, providers of intervention, target population, intervention, measurement and psychological outcome (Full description in Annex 3).

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232 Table 3 Characteristics of the reviewed Randomised controlled trial (RCT) studies, intervention, measurement and psychological outcome

Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Significance				
					Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Dincer and Inangil, Turkey(33)	Certified emotional freedom technique	Nurses	Activating and deactivating signals to brain by stimulating points on skin	N = 72 Ni = 35 Nc = 37	+				
Fiol-DeRoque et al., Spain(34)	Self-conducted (online application)	Health care workers	PsyCovidApp, targeting emotional skills, healthy lifestyle behaviour, burnout, and social support	N = 436 Ni = 248 Nc = 234	NoAssoc	NoAssoc	NoAssoc	NoAssoc	
Guan, et al. (Study 2B), China(44)	Self-conducted (online)	General population	Self-compassion writing task	N = 79 Ni = 40 Nc = 39	+				
Guan et al. (Study 3), China(44)	Self-conducted (online)	College students	Self-compassion meditation and writing tasks	N = 95 Ni = 50 Nc = 45	+				
Kahlon et al., US(40)	Trained callers (Telephone-based)	Homebound older adults and people with disabilities	Phone call with empathetic conversations	N = 240 Nc = 120 Ni = 120	+		+		
Khademi et al., Iran(27)	Self-conducted	COVID-19 patients	Performed mandala colouring	N = 70 Ni = 35 Nc = 35	+				
Liu et al., China(48)	Trainer	COVID-19 patients	Jacobson's relaxation techniques	Intervention = 51 Ni = 25 Nc = 26	+			+	
Malboeuf-Hurtubise et al., Canada)41(	Trained students in psychology (Video)	School students	Mindfulness-based interventions and philosophical discussions on COVID-19	N = 37 (Pre-post intervention assessment)	+				

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Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Significance				
					Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Parizad et al., Iran(28)	Nurses	COVID-19 Patients	Guided imagery session with five audio tracks	N = 110 Nc = 55 Ni = 55	+				
Shaygan et al., Iran)29(	Self-conducted (online multimedia)	COVID-19 patients	Cognitive behavioural techniques, stress management, mindfulness, and positive psychotherapy	N = 48 Ni = 26 Nc = 22		+			
Smith et al., US)42(	Self-conducted (online application)	Obstetric and gynaecology patients	Mobile meditation app: encouragement, mindfulness meditation, sleep stories, nature sounds	N = 101 Ni = 50 Nc = 51	+	+	+	+	
Solianik et al., Lithuania)38(	Tai chi instructor (online)	General population >60 years	Tai chi practice	N = 30 Ni = 15 Nc = 15	+	+	+		
Wadhen et al., UK(39)	Yoga teacher (online)	General population	Yoga class via zoom meetings	N = 34 Nc = 17 Ni = 17	NoAssoc	NoAssoc	+		
Zhang et al., China(50)	Self-conducted (Online)	Residents	Mindfulness-Based Stress Reduction program	N = 51 Ni = 29 Nc = 28	+	+	+		

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236 Table 4 Characteristics of the reviewed Non-randomised controlled study (NRS) studies, intervention, measurement and psychological outcome

Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Guan, et al. (Study 2A), China(44)	Self-conducted (online)	Young adults	A live session of self-compassion meditation	N = 64 (Pre-post intervention assessment)	+				
Giordano et al., Italy(30)	Trained music therapist	Clinical staff	Music therapy supplemented by guided imagery and music	N = 34 (Pre-post ntervention assessment)					+
González-García et al., Spain(35)	Mindfulness teacher and psychologist (online)	University students	Minilectures, mindfulness and compassion practices and self-reflection exercises	N = 66 (Pre-post intervention assessment)	+	+	+	+	
Gorbeña, et al., Spain(36)	Facilitators (interview and music playlists)	University students	Well-being and Personal Development Program	N = 151 Ni = 112 Nc= 39		+		+	
Kim et al., South Korea(45)	Psychiatrists (Telephone-based)	COVID-19 patients	30-minute telephone consultation providing information and support for COVID-19 infection and hospital isolation	Week1 n=33 Week2 n=19 (Pre-post intervention assessment)	±		±	±	±
Li et al., China(46)	Teacher (Video)	Nurses	Simulation training about using personal protective devices	N = 60 (Pre-post intervention assessment)	+		+		
Liu et al., China(47)	Self-conducted	Nurses	Diaphragmatic breathing relaxation training	N = 140 (Pre-post intervention assessment)	+		NoAssoc	+	
Luo et al., China(49)	Nurses	COVID-19 patients	Auricular point pressure	N = 84 Ni = 66 Nc = 18	+			+	

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Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Matiz et al., Italy(31)	Socio-health educator and psychologist	School teachers	Mindfulness-oriented meditation training program	N = 67 (Pre-post intervention assessment)	+		+		
Niu et al., China(51)	Nurses	(Suspected) COVID-19	Providing information about isolation, treatment, and social emotional support	N = 137 (Pre-post intervention assessment)	+		+		
Philip et al., France(32)	Self-conducted (online application)	General population	KANOPEE app with virtual agent interaction	N = 47 (Pre-post intervention assessment)				+	
Riva et al. Italy)37(	Self-conducted	General population	360◦ VR video entitled 'The Secret Garden' and a series of social exercises	N = 38 (Pre-post intervention assessment)	NoAssoc	+	+		
Sylvia et al., US(43)	Psychologists, physicians, social workers, nurses (online)	Health care workers	Stress management and resilience training, relaxation response program	N = 102 (Pre-post intervention assessment)	+		+		
Tunuguntla et al., Global level(26)	Self-conducted (online)	General population	Yoga and meditation with sound therapy	N = 820 (Pre-post intervention assessment)				+	
Yang et al., China(52)	Psychotherapist and nurse (in-person and online)	COVID-19 patients	Psychotherapy, empathy, muscle and breath relaxation, and cognitive behavioural therapy	N = 35 (Pre-post intervention assessment)	+		+	+	

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Note:
+ denotes Statistically significant association with post-intervention mental health improvements.
NoAssoc denotes No association between measure and post-intervention symptoms.
± denotes mixed results of association and no association, due to difference in timing of outcome measurement
Grey shade denotes not applicable.

Of 24 anxiety interventions from 18 publications, 21 (87.5%) significantly lowered anxiety in intervention groups; 11 were RCT(27,28,33,38,40–42,44,48,50) and 10 quasi-experimental studies(31,35,43–47,49,51,52). Among the 11 RCT studies, most interventions were mindfulness and meditation-based(38,41,42,44,50). Two studies applied specific techniques such as Emotional Freedom Techniques(33) and progressive muscle relaxation and deep breathing techniques(48).

Fourteen studies provided different interventions to reduce depression, of which 11 (78.6%) significantly reduced depression(31,37–40,42,43,46,50–52). Of 14 studies, five of them were RCT; four related to mindfulness and meditation-based stress reduction(42,50), focusing on mind-body exercises including yoga(39) and tai chi(38)and one provided empathetic conversations through phone calls(40).

Of nine studies on insomnia, seven found that the interventions significantly reduced insomnia or improved sleep quality (77.8%)(26,32,42,47–49,52). Of these nine studies, only two were RCT, with one using a mobile meditation application(42), and the other using progressive muscle relaxation and deep breathing techniques(48).

Of total nine interventions against stress, seven had shown significant positive outcomes in stress reduction (77.8%) (29,35–38,42,50). One study shows no association of interventions and outcome(39), while one study showed positive and negative results due to timing of assessment(45). Three studies reported the effectiveness of interventions on self-compassion; participants reported significantly higher perception of self-compassion than those in the control group(35,43,44). One study using pre- and post- assessments show improvement of loneliness(40). Only one study examined suicide, 30-minute telephone consultation was provided which was effective after one week.)(45.

Risk of bias

The quality of reporting for 14 RCT studies was assessed using the Cochrane risk of bias tool. Table 5 shows the risk of bias by six domains. These RCT studies had no risk of attrition bias (incomplete outcome data) or reporting bias (selective reporting). Of 14 studies, five (35.7%) had low risk of bias for all six dimensions reflecting high quality RCT. Most studies had unclear risk on allocation concealment, random sequence generation and blinding participants and personnel(27,38,41,44,48). Two studies (14.3%) reported high risk of bias on blinding of participants and personnel(39,50).

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Table 5 Risk of bias assessments for all 14 RCT studies

	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Incomplete outcome data	Selective reporting	Other bias	Number of low risk out of six criteria
Dincer and Inangil (33)	Low	Low	Low	Low	Low	Low	6
Fiol-DeRoque et al.(34)	Low	Low	Low	Low	Low	Low	6
Guan et al. (study 2b)(44)	Unclear	Unclear	Unclear	Low	Low	Low	3
Guan et al. (study 3)(44)	Unclear	Unclear	Unclear	Low	Low	Low	3
Kahlon et al.(40)	Low	Low	Low	Low	Low	Low	6
Khademi et al.(27)	Low	Unclear	Unclear	Low	Low	Low	4
Liu et al.(48)	Unclear	Unclear	Unclear	Low	Low	Low	3
Malboeuf-Hurtubise et al.)4:	Unclear	Unclear	Low	Low	Low	Unclear	3
Parizad et al.(28)	Low	Low	Low	Low	Low	Low	6
Shaygan, et al.)29(	Low	Low	Low	Low	Low	Low	6
Smith, et al.)38,42(	Low	Low	Unclear	Low	Low	Low	5
Solianik, et al.)38(	Unclear	Unclear	Low	Low	Low	Low	4
Wadhen, et al.(39)	Low	Unclear	High	Low	Low	Low	4
Zhang, et al.(50)	Low	Unclear	High	Low	Low	Low	4

DISCUSSION

Study characteristics

All 29 studies in this review were carried out in 2020 in all geographical continents except Africa and Australia. The majority of studies were from high income countries. Therefore, the varieties of interventions in this review can be useful for adaptive application in different country contexts, although there is no study from low-income countries. Psychological impacts are caused not only by direct consequences of the pandemic(53–55), but also by public health and social containment measures used by governments, notably movement restrictions and lockdowns(56). Over a half of the psychological interventions reviewed in this study were delivered to COVID-19 patients, health care workers and general population via an online platform, as on-site intervention was not possible in the pandemic context. The psychological consequences from this pandemic are diverse, requiring a wide range of interventions and tools for measuring subsequent psychological outcomes, including suicidal ideation, depression, anxiety, stress, and insomnia.

Each outcome was evaluated using a variety of tools. The BDI was the most commonly used to assess suicidal ideation, followed by the HADS for screening depression, the STAI for screening anxiety, the PSS for screening stress, and the ISI for screening insomnia. The measures were all used for screening rather than diagnosis. Several measures were validated, with Cronbach's alpha values greater or equal to 0.7 indicating high reliability . The majority of the tools were self-assessment questionnaires that participants in the study could complete using the sample scoring system. However, no articles reported on the sensitivity and specificity of each measure**.**

Interventions and psychological outcomes

Overall, most interventions have positive impact in mitigating psychological consequences in COVID-19 patients, health care workers and general population. A singular intervention can have combined effects for multiple psychological outcomes; for example one RCT using mobile meditation

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application including verbal and written encouragement, mindfulness meditation, sleep stories, nature sounds has significant positive effects on reduction of depression, anxiety, stress and insomnia(42). However, another study using an online application targeting emotional skills, healthy lifestyle behaviour, burnout, and social support reported no significant difference in the same measured outcomes(34).

Almost all 14 RCT studies that included mindfulness and meditation interventions to reduce depression, anxiety and stress showed significantly improved outcomes (depression(38–40,42,50), anxiety(27,28,33,38–42,44,50), stress(38,42,50)) with the exception of one study (39), which did not improve anxiety. Stress, anxiety, and depression is caused by sympathetic nervous system activation due to widespread depolarization throughout the brain and body; on the other hand, meditation and deep breathing lead to parasympathetic nervous system activation due to widespread inhibition and hyperpolarization(58). Recently, mindfulness and meditation-based interventions have increasingly been integrated into mental health interventions(59); meditation practices facilitate training of the self-regulation of attention and awareness, thereby enhancing control of mental processes, and consequently increasing wellbeing. Meditation and breathing techniques such as yoga, tai-chi, targeting body and mind are also applied and are effective in reducing stress, anxiety, and depression(60,61). Yoga, for example, helps with intellectual and mental exercise, improves the psychological management and monitoring of stress and negative emotions, and supports mental balance(61–63). Mindfulness and meditation-based interventions are readily available, and do not pose threats or risks. It can be applied as a first-line or complementary intervention for stress, anxiety, depression, and some emotional disorders during COVID-19.

Moreover, combining mindfulness intervention with online application technology can benefit people who are quarantined or in isolation wards with no direct access to psychotherapy. In this review, online multimedia psychoeducational interventions were demonstrated to significantly reduce stress(29); mindfulness-based stress reduction, cognitive–behavioural techniques, stress management techniques, and positive psychotherapy have been delivered via WhatsApp. Nowadays, internet-based, self-administrated multimedia and education can promote access to mental health interventions, particularly amidst COVID-19. Online-based interventions enable people to social distance and be time-efficient and cost-effective.

The pandemic does not allow in-person on-site intervention, causing a major shift to remote care(64–66) and telemedicine; the use of electronic communication to deliver health care services at a distance, becomes a key supporting measure for health services delivery systems during this pandemic(67,68). Several channels of telemedicine are accessible by patients and health professionals including telephone calls, video calls, interactive mobile health applications (m-Health), short message service, e-mail, secure remote patient monitoring, and video conferencing. Video calls via social media is the most common channel of communication, but special communication softwares (e.g., Zoom) or current messenger applications (e.g., WhatsApp, and WeChat) are popular and convenient options as well(69).

Furthermore, several countries have developed digital health technology to facilitate health care delivery through an online platform(70,71). Several challenges remain; online interventions cannot replace in-person therapy, notably when there is high risk of suicide which necessitates the presence of a psychotherapist or human interaction(72,73). Furthermore, several online counsellors(74) raised concerns from their experiences, such as a lack of non-verbal language for interpreting patients who struggle to verbally communicate their feelings(75) or significantly shorter counseling periods through the telephone due to “more superficial responses to questions”(76). However, the pandemic has raised awareness for the necessity of the online interventions where limitations can be resolved, as several studies have reported benefits of online services(77–80).

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Bias assessment

Most common reason across items of bias assessment was related to incomplete or unclear reporting in the RCT (81). Between 2011-2014, 24.7 % of the RCTs included in the Cochrane Review demonstrated an unclear risk of bias in the domain of incomplete outcome data (82). But all 14 RCT in the review presented complete outcome data and reported loss to follow-up for risk of attrition bias. In this review, each article was assessed by at least two authors and we compared the risk of bias assessment for each item in terms of 'high', 'low' or 'unclear' risk of bias between the two reviews.

More than half of the studies had unclear risk on allocation concealment due to the method of concealment was not described or inadequate detail to allow a clear judgment. Similarly, 50.1% of reported the April 2011 issue of the Cochrane Database of Systematic Reviews had unclear risk of bias on allocation concealment (83). Concealment of randomized allocation prevents an influence of patient characteristics on allocation to intervention and control groups. One study found that lack of adequate allocation concealment was associated with overestimation of treatment effect (84).

Two studies reported high risk of bias on blinding of participants and personnel (39,50). Blinding of participants and personnel prevents differences in patient management between groups and blinding of outcome assessors also prevents knowledge of the assigned intervention group influencing outcome measurement. However, both studies were conducted through online platform and the group memberships were not blind.

Limitations

Causal interpretations are limited by 15 non-randomised control study design. In addition to variation in sample size across studies, there were also variations in the mode of delivery of interventions, participant characteristics, and measurement tools across different study sites. Furthermore, a majority of the trials included small sample sizes which is commonplace for studies on psychological interventions. High heterogeneity of selected studies led to inability to conduct meta-analysis. There was also a lack of assessment of co-interventions, which provide external sources of support to the participants in the intervention and control groups and may have influenced results of the outcomes of an intervention. Lastly, post-intervention outcomes from the studies were measured from short follow-up durations, so it is not possible to conclude the long-term psychological effects of those interventions

CONCLUSION AND POLICY IMPLICATIONS

Of the total reviewed 29 studies, 26 diagnostic or screening measures were applied; eight for anxiety, seven for depression, five for stress, five for insomnia and one for suicide. Most studies conducted in high income and upper-middle income countries. Half of studies targeted general population and the remaining in COVID-19 patients and health care workers. Among 14 RCT studies, most interventions were effective in reducing psychological consequences.

Despite study limitations, the use of measurement measures and the strength of evidence from this review suggests an overall positive impact of the different interventions in terms of reducing psychological consequences in the population and health care workers. These effective interventions can be replicated and applied in other country settings, for which modes of delivery can be adjusted in line with country resources, epidemic situation, and social and health systems contexts. Moreover, it is necessary to improve the use of evidence in psychological interventions in mental health services.

We recommend that these effective interventions can be replicated and applied in other country settings, for which modes of delivery can be adjusted in line with country resources, epidemic

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situation, and social and health systems contexts. In the first quarter of 2022, the pandemic is still far from over due to low immunization coverage in African countries and continued emergence of Variants of Concerns in the context of pressure to resuming the economic activities and opening the country borders. In ending the acute phase of pandemic, rapid scale up and promote vaccine acceptance are key policy actions.

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Annex 1: Search term and results

1) PUBMED = 1,564
- COVID* OR novel corona* OR nCoV* OR 2019nCoV OR severe acute respiratory syndrome coronavirus 2 OR severe acute respiratory pneumonia outbreak OR SARS CoV 2 OR SARS-CoV-2 (206,274)
- therap*[MeSH Terms] OR intervention[MeSH Terms] OR consultation[MeSH Terms] (7,081,376)
- Mental Health [MeSH terms] OR Mental Disorders [MeSH terms] OR Mental Illness [MeSH terms] OR Suicide OR Depression OR Anxiety OR Psychological Distress [MeSH terms] OR Emotional Distress [MeSH terms] OR Insomnia [MeSH terms] (1,846,124)

2) EMBASE = 1,114
- COVID* (sh) or novel corona*.mp. or nCoV*.mp. or 2019nCoV.mp. or severe acute respiratory syndrome coronavirus 2 (sh) or severe acute respiratory pneumonia outbreak (sh) or SARS CoV 2.mp. or SARS-CoV-2.mp. (81,488)
- (treatment or therap*).mp. or intervention (sh) or consultation (sh) (13,205,742)
- (Mental Health or Mental Disorders or Mental Illness).mp. or Suicide (sh) or Depression (sh) or Anxiety (sh) or Psychological Distress.mp. or Emotional Distress.mp. or Insomnia (sh) (1,017,858)

3) PsycInfo = 1,524
- DE "Coronavirus" OR DE "COVID-19" OR DE "Severe Acute Respiratory Syndrome" (6,545)
- intervention or program or counseling (1,095,058)

4) MEDLINE(OVID) = 1,417
- (COVID* or novel corona* or nCoV* or 2019nCoV or severe acute respiratory syndrome coronavirus 2 or severe acute respiratory pneumonia outbreak or SARS CoV 2 or SARS-CoV-2).mp. (124,422)
- (therap* or therap* or intervention or consultation).mp. (6,269,680)
- (Mental Health or Mental Disorders or Mental Illness or Suicid* or Depression or Anxiety or Psychological Distress or Emotional Distress or Insomnia).mp. (638,321)

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Annex 2

Table A2 Tools for the assessment of psychological consequences with score classification

Outcomes	Tools	Method description	Score Classification	Studies
1. Suicidal Ideation Assessment	1.1 Beck Depression Inventory (BDI) (item 9)	A 21-item, self-reported rating inventory that measures attitudes and symptoms of depression. Item-9 of the test asked the potential to kill oneself (49).	Score 1-10: Normal Score 11-16: Mild Mood Disturbance Score 17-20: Borderline Clinical Depression Score 21-30: Moderate Depression Score 31-40: Severe Depression Score over 40: Extreme Depression	Kim et al., 2020(44)
2. Depression Assessment	2.1 Patient Health Questionnaire-9 (PHQ-9)	A 9-item assessment that is used to assess the frequency the patients experience depressive symptoms according to DSM-IV depression criteria(50).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score 15-19: Moderately Severe Score 20-27: Severe	Niu et al., 2021(50) Yang et al., 2020(51)
	2.2 Patient Health Questionnaire for Depression-8 (PHQ-8)	An 8-item assessment that consists of 8 of the 9 criteria on which the DSM-IV diagnosis of depressive disorders is based (51).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score 15-19: Moderately Severe Score 20-24: Severe	Kahlon et al., 2021(39)
	2.3 Patient Health Questionnaire-4 (PHQ-4)	A 4-item questionnaire that aims to allow for brief and accurate measurement of core symptoms or signs of depression and anxiety (52).	Score 0-2: Normal Score 3-5: Mild Score 6-8: Moderate Score 9-12: Severe	Sylvia et al., 2021(42)
	2.4 Depression Anxiety Stress Scale (DASS-21)	A 21-item test, short version of the original self-report questionnaire (DASS-42 item) to evaluate depression, anxiety, and stress symptoms with 7 items per subscale; (1) DASS-21 Depression, (2) DASS-21 Anxiety, and (3) DASS-21 Stress (53).	**Depression** Score 0-9: Normal Score 10-13: Mild Score 14-20: Moderate Score 21-27: Severe Score over 28: Extremely Severe **Anxiety** Score 0-7: Normal Score 8-9: Mild Score 10-14: Moderate Score 15-19: Severe	Riva et al., 2021(36), Wadhen et al., 2021(38), Fiol-DeRoque et al., 2021(33)

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Outcomes	Tools	Method description	Score Classification	Studies
			Score over 20: Extremely Severe **Stress** Score 0-14: Normal Score 15-18: Mild Score 19-25: Moderate Score 26-33: Severe Score over 34: Extremely Severe	
	2.5 Hospital Anxiety and Depression Scale (HADS)	A 14-item assessment used to measure anxiety and depression symptoms in medical patients. The HADS produces two scales with 7 items in each scale; one for anxiety (HADS–A) and one for depression (HADS–D) (54).	**Depression and Anxiety** Score 0-7: Normal Score 8-10: Borderline abnormal (borderline case) Score 11-21: Abnormal	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (HADS-D)(44)
	2.6 Self-rating Depression Scale (SDS)	A 20-item assessment to assess the subjective symptoms of depression for up to one week(55).	Score <50: Normal Score 50-59: Mild Depression Score 60-69: Moderate to Marked Major Depression Score >70: Extreme Major Depression	Liu et al., 2021(46), Li et al., 2020(45)
	2.7 Brief Symptom Inventory 18 (BSI-18)	An 18-item questionnaire, taken from the 53-item Brief Symptom Inventory (BSI). The scale includes three subscales (somatization, depression, and anxiety) (56).	The interpretation of the BSI-18 uses Global Severity Index (GSI) of distress represents sum across the 3 subscales range from 0-72. Higher scores indicate higher levels of psychological distress.	Zhang et al., 2021(49)
3. Anxiety assessment	3.1 Spielberger State-Trait Anxiety Inventory (STAI)	A 40-item self-report tool consisting of two parts for measuring state and trait anxiety with 20 questions in each section(57).	Score 20-31: Mild Score 32-42: Mild to Moderate Score 43-53: Moderate to Severe Score 54-64: Relatively Severe Score 65-75: Severe Score ≥ 76: Very Severe	Parizad et al., 2021(27), Khademi et al., 2021(26), González-García et al., 2021(34), Liu et al., 2020(47), Dincer et al., 2021(32), Guan et al., 2021(43)

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Outcomes	Tools	Method description	Score Classification	Studies
	3.2 Anxiety - Generalized Anxiety Disorder scale (GAD-7)	A 7-item questionnaire to detect symptoms of generalized anxiety disorder based on seven core symptoms and assess the frequencies with which respondents experience these symptoms within the prior 2 weeks (58).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score >=15: Severe	Kahlon et al., 2021(39), Luo et al., 2021(48), Niu et al., 2021(50), Yang et al., 2020(51)
	3.3 Self- Rating Anxiety Scale (SAS)	A 20-item assessment evaluating a variety of anxiety symptoms, including both psychological and somatic symptoms, that respondents experience within the past month (59).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score >=15: Severe	Liu et al., 2021 (46), Guan et al., 2021(43), Li et al., 2020(45)
	3.4 Behavior Assessment Scale for Children-3rd edition, (BASC III)	A test to identify and manage behavioral and emotional strengths and weaknesses of children and adolescents from three perspectives: teacher, parent and self (starting at age 8) (60).	Interpretation using the mean score. Higher scores indicate higher levels of anxiety.	Malboeuf-Hurtubise et al., 2021(40)
	3.5 Patient Health Questionnaire-4 (PHQ-4)	See 2.3		Sylvia et al., 2021 (42)
	3.6 Depression Anxiety Stress Scale (DASS-21)	See 2.4		Riva et al., 2021(36), Fiol-DeRoque et al., 2021(33), Wadhen et al., 2021(38)
	3.7 Hospital Anxiety and Depression Scale (HADS)	See 2.5	Score 20-44: Normal Score 45-59: Mild to Moderate Anxiety Score 60-74: Marked to Severe Anxiety levels	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (44)
	3.8 Brief Symptom Inventory 18 (BSI-18)	See 2.7		Zhang et al., 2021 (49)

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Outcomes	Tools	Method description	Score Classification	Studies
4. Psychological distress and stress assessment	5. The General Health Questionnaire (GHQ-12)	A 12-item test, self-administered to evaluate mental distress and common mental disorders. There were multiple versions of GHQ including GHQ-60, GHQ-30, GHQ-28, and GHQ-12 (61).	The average sum of its items provides a scalar indicator of the degree of mental distress. Score of 11 or 12: typical Score >15: Suggesting evidence of distress Score >20: Severe problems with psychological distress	Gorbeña et al., 2021 (35)
	4.1 Perceived stress 14-item (PSS-14)	A 14-item scale used as a psychological instrument to measure the perception of stress, with seven positive items and seven negative items (62).	Score 0-18: Low Stress Score 19-37: Moderate Stress Score 38-56: High Stress	Wadhen et al., 2021(38), Shaygan et al., 2021 (28)
	4.2 Perceived Stress Scale 10-item (PSS-10)	A 10-item questionnaire about feelings and thoughts during the last month(63) .	Score 0-13: Low Stress Score 14-26: Moderate Stress Score 27-40: High Perceived Stress	Riva et al., 2021(36), Solianik et al., 2021(37), Smith et al., 2021(41), González-García et al., 2021(34)
	4.3 Depression Anxiety Stress Scale (DASS-21)	See 2.4	Score 0-13: Low Stress Score 14-26: Moderate Stress Score 27-40: High Perceived Stress	Fiol-DeRoque et al., 2021(33)
	4.4 Brief Symptom Inventory 18 (BSI-18)	See 2.7		Zhang et al., 2021 (49)
5. Insomnia assessment	5.1 Insomnia Severity Index (ISI)	A 7-item assessment to assess current sleep characteristics(64).	Score 0-7: No clinically significant insomnia Score 8-14: Subthreshold Insomnia Score 15-21: Clinically Significant Insomnia (Moderate) Score 22-28: clinically Significant Insomnia (Severe)	Tunuguntla et al., 2021(25), Philip et al., 2020(31), Fiol-DeRoque et al., 2021(33), Kim, et al., 2020(44)
	5.2 St. Mary's Hospital Sleep Questionnaire (SMH)	A 14-item test asking about patient's sleep experience and comprises items to demonstrable reliability, with some open-ended questions(65).	The test needs to be evaluated by experts.	Luo et al., 2021 (48)

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Outcomes	Tools	Method description	Score Classification	Studies
	5.3 PROMIS (Patient-Reported Outcomes Measurement Information System) Sleep Disturbance Short Form	Items focus on perceptions of sleep quality, sleep depth, and restoration associated with sleep, perceived difficulties with getting to sleep or staying asleep, and perceptions of the adequacy of and satisfaction with sleep (66).	Score <55: None to slight Score 55-59.9: Mild Score 60-69.9: Moderate Score >70: Severe	Smith et al., 2021 (41)
	5.4 Pittsburgh Sleep Quality Index (PSQI)	An 18-item index measuring seven sleep dimensions (67).	The overall PSQI score ranges from 0 to 21, with higher scores reflecting poorer sleep quality. A revised PSQI score of 7 or higher is considered poor sleep quality. If the score of any individual dimension is higher than 1, there is a problem with that specific dimension.	Liu et al., 2021(46), Yang et al., 2020(51)
	5.5 Sleep State Self-Rating Scale (SRSS)	A 10-item assessment used to assess sleep state.	SRSS over or equal to 23 points is considered to have sleep problems.	Liu et al., 2020(47)

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Annex 3

Table A3 Characteristics of the reviewed studies, intervention, measurement and psychological outcome

Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Dincer and Inangil, Turkey(32)	RCT	Certified emotional freedom technique	Nurses	Activating and deactivating signals to brain by stimulating points on skin	N = 72 Ni = 35 Nc = 37	Anxiety	STAI (+)
Fiol-DeRoque et al., Spain(33)	RCT	Self-conducted (online application)	Health care workers	PsyCovidApp, targeting emotional skills, healthy lifestyle behaviour, burnout, and social support	N = 436 Ni = 248 Nc = 234	Depression Anxiety Stress Insomnia	PHQ-8 (na) GAD7 (na) DASS-21 (na) ISI (na)
Guan, et al. (Study 2B), China(43)	RCT		General population	Self-compas sion writing task	N = 79 Ni = 40 Nc = 39	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Guan et al. (Study 3), China(43)	RCT		College students	Self-compassion meditation and writing tasks	N = 95 Ni = 50 Nc = 45	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Kahlon et al., US(39)	RCT	Trained callers (Telephone-based)	Homebound older adults and people with disabilities	Phone call with empathetic conversations	N = 240 Nc = 120 Ni = 120	Depression Anxiety Loneliness	GAD7 (+) GAD7 (+) UCLA Loneliness Scale (+), De Jong Loneliness Scale (na)
Khademi et al., Iran(26)	RCT	Self-conducted	COVID-19 patients	Performed mandala colouring	N = 70 Ni = 35 Nc = 35	Anxiety	GAD-7 (+)
Liu et al., China(47)	RCT	Trainer	COVID-19 patients	Jacobson's relaxation techniques	Intervention = 51 Ni = 25	Anxiety Insomnia	STAI (+) SRSS (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					Nc = 26		
Malboeuf-Hurtubise et al., Canada(40)	RCT	Trained students in psychology (Video)	School students	Mindfulness-based interventions and philosophical discussions on COVID-19	N = 37 (Pre-post intervention assessment)	Anxiety	BASC III (+)
Parizad et al., Iran(27)	RCT	Nurses	COVID-19 Patients	Guided imagery session with five audio tracks	N = 110 Nc = 55 Ni = 55	Anxiety	STAI (+)
Shaygan et al., Iran(28)	RCT	Self-conducted (online multimedia)	COVID-19 patients	Cognitive behavioural techniques, stress management, mindfulness, and positive psychotherapy	N = 48 Ni = 26 Nc = 22	Stress	PSS (+)
Smith et al., US(41)	RCT	Self-conducted (online application)	Obstetric and gynaecology patients	Mobile meditation app: encouragement, mindfulness meditation, sleep stories, nature sounds	N = 101 Ni = 50 Nc = 51	Depression Anxiety Stress Insomnia	HADS (+) HADS (+) PSS (+) PROMIS (+)
Solianik et al., Lithuania(37)	RCT	Tai chi instructor (online)	General population >60 years	Tai chi practice	N = 30 Ni = 15 Nc = 15	Depression Anxiety Stress	HADS (+) HADS (+) PSS-10 (+)
Wadhen et al., UK(38)	RCT	Yoga teacher (online)	General population	Yoga class via zoom meetings	N = 34 Nc = 17 Ni = 17	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (na), PSS (na)
Zhang et al., China(49)	RCT	Self-conducted (Online)	Residents	Mindfulness-Based Stress Reduction program	N = 51 Ni = 29	Depression Anxiety	BSI-18 (+) BSI-18 (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					Nc = 28	Stress	BSI-18 (+)
Kim et al., South Korea(44)	NRS	Psychiatrists (Telephone-based)	COVID-19 patients	Ward telephone providing information and support for COVID-19 infection and hospital isolation	Week1 n=33 Week2 n=19 (Pre-post intervention assessment)	Suicidal idea Depression Anxiety Insomnia	BDI: week1 (+), week2 (na) HADS: week1 (+), week2 (na HADS: week 1 (+), week 2 (na) ISI: week 1 (+), week 2 (na)
Guan, et al. (Study 2A), China(43)	NRS	Self-conducted (online)	Young adults	A live session of self-compassion meditation	N = 64 (Pre-post intervention assessment)	Anxiety	PHQ-4 (+)
Giordano et al., Italy(29)	NRS	Trained music therapist	Clinical staff	Music therapy supplemented by guided imagery and music	N = 34 (Pre-post intervention assessment)	Others: fear/ worry	MTC (+)
González-García et al., Spain(34)	NRS	Mindfulness teacher and psychologist (online)	University students	Minilectures, mindfulness and compassion practices and self-reflection exercises	N = 66 (Pre-post intervention assessment)	Anxiety Stress Self-compassion	DASS-21 (+) DASS-21 (+) DASS-21 (+)
Gorbeña, et al., Spain(35)	NRS	Facilitators (interview and music playlists)	University students	Well-being and Personal Development Program	N = 151 Ni = 112 Nc= 39	Stress	PSS (+)
Li et al., China(45)	NRS	Teacher (Video)	Nurses	Simulation training about using personal protective devices	N = 60 (Pre-post intervention assessment)	Depression Anxiety	SDS (+) SAS (+)
Liu et al., China(46)	NRS	Self-conducted	Nurses	Diaphragmatic breathing relaxation training	N = 140 (Pre-post	Depression	SDS (na)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					intervention assessment)	Anxiety Insomnia	SAS (+) PSQI (+)
Luo et al., China(48)	NRS	Nurses	COVID-19 patients	Auricular point pressure	N = 84 Ni = 66 Nc = 18	Anxiety Insomnia	GAD-7 (+) SMH score (+)
Matiz et al., Italy(30)	NRS	Socio-health educator and psychologist	School teachers	Mindfulness-oriented meditation training program	N = 67 (Pre-post intervention assessment)	Depression Anxiety	HADS (+) HADS (+)
Niu et al., China(50)	NRS	Nurses	(Suspected) COVID-19	Providing information about isolation, treatment, and social emotional support	N = 137 (Pre-post intervention assessment)	Depression Anxiety	PHQ-9 (+) GAD-7 (+)
Philip et al., France(31)	NRS	Self-conducted (online application)	General population	KANOPEE app with virtual agent interaction	N = 47 (Pre-post intervention assessment)	Insomnia	ISI (+)
Riva et al. Italy(36)	NRS	Self-conducted	General population	360◦ VR video entitled 'The Secret Garden' and a series of social exercises	N = 38 (Pre-post intervention assessment)	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (+), PSS (+)
Sylvia et al., US(42)	NRS	Psychologists, physicians, social workers, nurses (online)	Health care workers	Stress management and resilience training, relaxation response program	N = 102 (Pre-post intervention assessment)	Depression Anxiety Self-compassion	PHQ-4 (+) PHQ-4 (+) SCS (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Tunuguntla et al., Global level(25)	NRS	Self-conducted (online)	General population	Yoga and meditation with sound therapy	N = 820 (Pre-post intervention assessment)	Insomnia	ISI (+)
Yang et al., China(51)	NRS	Psychotherapist and nurse (in-person and online)	COVID-19 patients	Psychotherapy, empathy, muscle and breath relaxation, and cognitive behavioural therapy	N = 35 (Pre-post intervention assessment)	Depression Anxiety Insomnia	PHQ-9 (+) GAD-7 (+) PSQI (+)

RCT: Randomised controlled trial

NRS: Non-randomised controlled study

+ = Statistically significant association with post-intervention mental health improvements.

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4, Line 73-81
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5, Line 87-90
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Table 1
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P 5, Line 100
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Figure 1
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P 5, Line 106
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Table 1
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P9, Line 135
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P9, Line 124
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Table 1
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	NA
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	P9, Line 124

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P9, Line 141
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	P9, Line 141, Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 4 and 5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table 4
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Table 6
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P27, Line 329
	23b	Discuss any limitations of the evidence included in the review.	P27, Line 380
	23c	Discuss any limitations of the review processes used.	P27, Line 380
	23d	Discuss implications of the results for practice, policy, and future research.	P27, Line 390
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Not registered
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P30, Line 402

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Competing interests	26	Declare any competing interests of review authors.	NA
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Living through the psychological consequences of COVID-19 pandemic: a systematic review of effective mitigating interventions

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Title: Living through the psychological consequences of COVID-19 pandemic: a systematic review of effective mitigating interventions

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Structured abstract 301 words
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Table 5
Figure 1
Reference 85
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ABSTRACT

Objective

This review assesses interventions and their effectiveness in mitigating psychological consequences from pandemic.

Method

Published English literature were searched from four databases [Medline, PubMed, Embase, and Psycinfo] from January 2020 and September 2021. Total 27 papers with 29 studies (one paper reported three studies) met inclusion criteria. Cochrane risk of bias tool is applied to assess quality of all randomized control trials (RCT).

Results

All studies were recently conducted in 2020. Publications were from by high- (13, 44.8%), upper-middle (12, 41.4%) and lower-middle income countries (3, 10.3%) and global (1, 3.5%). Half of the studies conducted for general population (51.7%). One-third of studies (27.6%, 8) provided interventions to COVID-19 patients and 20.7% to health care workers. Of 29 studies, 14 studies (48.3%) were RCT. All RCT were assessed for risk of biases; five studies (15, 35.7%) had low risk as measured against all six dimensions reflecting high quality study.

Of these 29 studies, 26 diagnostic or screening measures were applied; eight (30.9%) for anxiety, seven (26.9%) for depression, five (19.2%) for stress, five (19.2%) for insomnia and one (3.8%) for suicide. Measures used to assess the baseline and outcomes of interventions were standardized and widely applied by other studies with high level of reliability and validity. Of 11 RCT studies, ten studies (90.9%) showed that anxiety interventions significantly lowered anxiety in intervention groups. Five of the six RCT studies (83.3%) had significantly reduced level of depression. Most interventions for anxiety and stress were mindfulness and meditation-based.

Conclusions

Results from RCT studies (11%, 78.6%) were effective in mitigating psychological consequences from COVID-19 pandemic when applied to health care workers, COVID-19 patients and general population. These effective interventions can be applied and scaled up in other country settings through adaptation of modes of delivery suitable to country resources, pandemic and health systems context.

Article summary

Strengths and limitations of this study

- Effective interventions which targeted patients, healhtcare workers and general population can be applied and scaled up by other countries.
- The heterogeneity of included studies such as the mode of delivery of interventions, participant characteristics, and measurement tools prevents efforts to conduct meta-analysis.
- There is a lack of assessment of the effects of co-interventions, provided to the participants in the intervention and control groups, which influence the outcomes in both groups.
- Measurement of outcome at short duration after interventions does not offer understanding of long-term outcome or its sustainability

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INTRODUCTION

Coronavirus disease, or COVID-19, was first reported in December 2019(1). By March 2020, the World Health Organization (WHO) declared the COVID-19 a pandemic and as of November 2021, there have been over 259 million cumulative confirmed cases and 5 million deaths from COVID-19 worldwide(2). The rapid global spread of disease has not only had physical health impacts, but also psychological consequences on the population.

The COVID-19 outbreak has had direct and indirect impacts on mental health. Many experienced fear and anxiety due to uncertainty about the pandemic's evolution, feelings of hopelessness, despair, and grief in the face of uncontrollable events(3,4). The impact of public health measures such as quarantine measures, social restrictions, and isolation have also resulted in mental health consequences including anxiety, depression, and feelings of neglect; anxiety was highly prevalent in people undergoing quarantine or social isolation(5–7). The pandemic's indirect effects, such as unemployment due to economic downturn and the death of family members from disease, has led to depression and suicide(8–10). Decreased population mobility and high rates of COVID-19 infection were found to be significantly related to an increased prevalence of major depression disorder, which compared to pre-pandemic, there were an additional 53.2 million cases of major depressive disorder and 76.2 million cases of anxiety disorders in 2020 (11).

Psychological consequences from the pandemic also developed more significantly in certain groups of people. Long hours of COVID-19 clinical services caused significant pandemic-related burnout for frontline health care workers worldwide(12–14). Fear of transmitting the infection to family members and increased demand for health services also caused stress and sleep problems among medical professionals(15–17).

Multiple mental health interventions have been implemented during previous epidemics such as Ebola and Middle East Respiratory Syndrome, particularly for health care workers experiencing higher levels of burnout and psychological consequences than others(18). Effective interventions like psychosocial and administrative support in workplaces have been shown to reduce the rates of emotional exhaustion in health care workers(18). Interventions in other population groups during Ebola outbreaks reduced depression, stress, and anxiety(19–21). Implementing interventions to alleviate mental health problems during the COVID-19 pandemic has proven to be difficult due to various social restrictions; face-to-face interventions seems not feasible where online interventions including telehealth and multimedia application were used as alternatives(22–24).

Though various interventions to mitigate psychosocial consequences from the COVID-19 pandemic have been launched, the measures used, their delivery channels and effectiveness on population groups are not well understood. This systematic review assesses interventions and measures used, and the effectiveness of interventions in mitigating psychological consequences on different populations so that effective interventions can be up-scaled accordingly. The psychological consequences affect patients who suffered from the illnesses and stress from quarantine and isolation, healthcare workers suffered from stressful long hour work and general population facing indirect effects of government measures such as lock down and limited social activities.

We propose three review questions. First, what interventions are used to mitigate psychological consequences in the population and in health care workers? Second, what measures are used for assessing intervention outcome? Third, what is the effectiveness of different interventions? Findings from this review can inform health professionals and governments of effective interventions to prevent or mitigate mental health problems from the COVID-19 pandemic.

METHODS

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Search strategy

We used PICO framework, a common tool used to to develop literature search strategies to ensure the comprehensive searches recommended by the Cochrane Collaboration(25). It is used in evidence-based practice to formulate healthcare related question. PICO elements include Population, Interventions, Comparisons and Outcomes. In this review, opulation are individuals in countries or areas affected by COVID-19; interventions are exposure to COVID-19 pandemic and the clinical or non-clinical interventions which mitigate the psychological impact from the pandemic; comparison of outcomes between case and control groups or before and after interventions; and outcome are mental consequences including suicide, depression, anxiety, stress, and insomnia/reduced sleep quality. See the full search strategy in Annex 1.

Published literatures were electronically retrieved from four databases including Medline, PubMed, Embase and Psycinfo. The eligibility criteria are English-language studies between January 2020 – September 2021where interventions were provided with measurable outcomes. Table 1 provides inclusion and exclusion criteria.

Table 1 Search terms, inclusion and exclusion criteria

	Inclusion and exclusion criteria
Population	Inclusion: Individuals in countries or areas affected by COVID-19 including patients, health care workers, women, minority groups Exclusion: ● Individuals in countries or areas not affected by the COVID-19. ● Individuals having prior psychological symptoms
Intervention/ exposure	Inclusion: ● Exposure to COVID-19 pandemic ● Both clinical and non-clinical interventions, provided by therapists, psychotherapists/counsellors, psychologists and qualified mental health staff including music and physical activity. Exclusion: Not any
Comparison	Inclusion: ● Comparing prevalence or degree of psychological consequences before and after interventions in the same group. ● Comparing prevalence or degree of psychological consequences in control and intervention groups. Exclusion: Not any
Outcome	Inclusion: Mental health consequences including suicide, depression, anxiety, stress, and insomnia/reduced sleep quality. Outcomes must be quantified as proportions/prevalence rate or defined categorically. Exclusion: Not any
Study design	Inclusion: Experimental research designs; ● Randomised controlled trials (RCTs) ● Controlled before and after studies (CBAs), with baseline and post-intervention measurement for both groups Exclusion:

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	• Type of publication: editorials, commentary, letters to the editor, reviews • Studies that did not have the outcomes measures specified in the protocol.

Selection of publications

The search terms were identified based on search strategy and applied to search in title, abstract, keyword and full-text. After searching, the duplicated studies or data were removed then the abstracts were reviewed; those were excluded for the following reasons with agreement in inclusion and exclusion criteria.

Figure 1 shows the PRISMA flow of this process.

<Figure 1>

Figure 1 PRISMA Flow diagram of the review process
Note * Some studies measure multiple outcomes

In the process of abstract screening, three researchers (AL, PP and AC) independently reviewed the abstracts. Consensus was reached if two researchers agreed based on eligibility criteria. In case of disagreement, the third researcher reviewed and decided.

Assessment the risk of bias

The Cochrane risk of bias tool (25) was applied to assess the methodological quality of 14 randomised controlled trials included by this systematic review. There are six domains of risk of bias: 1) Random sequence generation (selection bias), 2) Allocation concealment (selection bias), 3) Blinding of participants, personnel and outcome assessors, 4) Incomplete outcome data (attrition bias), 5) Selective reporting (reporting bias) and 6) Other bias(25). Three authors (AS, PP and AC) separately evaluated each domain and categorised the risk of bias into 'low risk', 'high risk', or 'unclear risk'. The results of assessment were showed in terms of number of 'low risk' of bias, which is six in total. Different criteria were used for making risk of bias judgements for each domain. Insufficient information reported in studies resulted in 'unclear risk' of bias judgment. When there were conflicting views, discussion among the three authors was convened to reach consensus.

Data extraction and synthesis

Relevant contents in all included articles were extracted and synthesised into three variables in an Excel spreadsheet: a) characteristic of study: author, year of publication, objective, country, study design, b) interventions: provider, participant, and c) outcomes: different types of psychological consequence, and measures used for assessment of psychological consequence.

Patient and public involvement

No patients were involved in this study.

RESULTS

Our search strategies identified a total of 5,623 records, of which 5,619 from the four databases and four from hand-searching from references of key articles. After abstract screening, 705 duplicated records were removed while 4,602 records were not relevant. The full text of the remaining 316 records were retrieved and reviewed for eligibility. We found that 289 publications were not relevant to the review questions. Finally, 27 articles comprised of 29 studies were included for systematic review.

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Study characteristics

Characteristics of 27 included articles are described in Table 2. One paper reported three studies (44), totalling 29 studies; all of which were conducted in 2020. Of the 29 studies, one was conducted globally (3.5%) (26) and the rest were categorised according to WHO geographical area. Three studies (10.3%) were conducted in the Eastern Mediterranean region (27–29), 10 (34.5%) in the European region (30–39), four (13.8%) in the Americas (40–43) and 11 (37.9%) in the Western Pacific (44–52). Applying the World Bank's income group categories, 13 (44.8%) studies were in high income (30–32,34–40,42,43,45), 12 (41.4%) in upper-middle income (33,41,44,46–52), and three (10.3%) in lower-middle income groups (27–29); there were no studies from the low income group.

On population, out of the total 29 studies, eight studies (27.6%) targeted COVID-19 patients (27–29,45,48,49,51,52), six (20.7%) among health care workers (30,33,34,43,46,47), and more than half (51.7%) of the studies were general population (26,31,32,35–42,44,50). Out of 29 studies, 14 (48.3%) were randomised control trials (RCT) (27–29,33,34,38–42,44(2b,3),48,50). The remaining 15 (51.7%) were non-RCT studies.

The interventions reported by the studies can be classified into three types: 19 online-based interventions (65.5%) (26,28–30,32,34,35,37,39–45,47,48), six on-site-based interventions (20.7%) (27,33,38,49–51) and four combined online and on-site interventions (13.8%) (31,36,46,52).

On outcome measurement, out of total 29 studies, only one study assessed suicide ideation (3.5%) (45), while 14 studies measured depression outcome (48.3%) (31,34,37–40,42,43,45–47,50–52), 24 studies measured anxiety (82.8%) (27,28,31,33–35,37–52), nine studies (31.0%) measured psychological stress(29,34–39,42,50), nine studies (31.0%) measured sleep quality (26,32,34,42,45,47–49,52), and six studies (20.7%) measured other outcomes such as loneliness and self-compassion (30,35,40,43,44). Table 2 shows characteristics of the studies.

Table 2 Characteristics of the 29 included studies

	Characteristics	Reference
Geographic area (WHO regions)		
- Eastern Mediterranean Region	3 (10.3%)	27-29
- European Region	10 (34.5%)	30-39
- Region of the Americas	4 (13.8%)	40-43
- Western Pacific Region	11 (37.9%)	44-52
- Global	1 (3.5%)	26
Income groups (World bank)		
- High-income economies	13 (44.8%)	30-32, 34-40, 42, 43, 45
- Upper-middle-income economies	12 (41.4%)	33, 41, 44, 46-52
- Lower-middle-income economies	3 (10.3%)	27-29
- Global	1 (3.5%)	26
Participants		
- COVID-19 patients (confirmed/suspected)	8 (27.6%)	27-29, 45, 48, 49, 51, 52
- Health care workers	6 (20.7%)	30, 33, 34, 43, 46, 47
- General population (included student and teacher)	15 (51.7%)	26, 31, 32, 35-42, 44, 50

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Study design		
- Randomised control trial	14 (48.3%)	27-29, 33, 34, 38-42, 44(2B,3), 48, 50
- Non-randomised control trial	15 (51.7%)	26, 30-32, 35-37, 43, 44(2A), 45-47, 49, 51, 52
Intervention		
- Online	19 (65.5%)	26, 28-30, 32, 34, 35, 37, 39–45, 47, 48
- Onsite	6 (20.7%)	27, 33, 38, 49–51
- Combined online and onsite	4 (13.8%)	31, 36, 46, 52
Psychological outcome		
- Suicidal idea	1 (3.5%)	45
- Depression	14 (48.3%)	31, 34, 37–40, 42, 43, 45–47, 50–52
- Anxiety	24 (82.8%)	27, 28, 31, 33–35, 37–52
- Stress	9 (31.0%)	29, 34–39, 42, 50
- Insomnia/ sleep quality	9 (31.0%)	26, 32, 34, 42, 45, 47–49, 52

*** There are 29 studies out of 27 papers

Measures used for the assessment of psychological consequences

Out of 29 studies, 20 measures were used to assess psychological outcomes. We categorized the measures based on the outcome they measured including suicidal ideation depression, anxiety, stress, and insomnia (Annex 2).

Suicidal ideation was assessed by using one measure, the Beck Depression Inventory (BDI) item 9, whereas depression and anxiety were assessed by using seven and eight measures respectively. The Hospital Anxiety and Depression Scale (HADS) was most used for the assessment of depression, while the Spielberger State-Trait Anxiety Inventory (STAI) was used by six studies (25.0%) on anxiety (27,28,33,35,44,48). Two different measures were used to assess both depression and anxiety, including the Patient Health Questionnaire-4 (PHQ-4), and HADS. The Brief Symptom Inventory-18 (BSI-18) and Depression Anxiety Stress Scale (DASS-21) were used to assess depression, anxiety, and psychological distress.

Stress was evaluated primarily using five measures (55.6%) in 9 studies, though the Perceived Stress Scale 10-item (PSS-10) was most commonly used. Among the other outcomes, insomnia assessments were distinctive in that they consisted of five tests, each of which measured the patients' sleep quality. The Insomnia Severity Index was widely used to determine sleep quality in four studies(26,32,34,45). There were several measures assessing outcomes beyond the focus of this study, one test for self-compassion and two tests for loneliness(35,40,43,44).

Interventions, measurements and effectiveness

Table 3 (RCT studies) and Table 4 (non-RCT studies) describes study design, providers of intervention, target population, intervention, measurement and psychological outcome (See full description in Annex 3).

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239 Table 3 Characteristics of the reviewed Randomised controlled trial (RCT) studies, intervention, measurement and psychological outcome

Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Significance				
					Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Dincer and Inangil, Turkey(33)	Certified emotional freedom technique	Nurses	Activating and deactivating signals to brain by stimulating points on skin	N = 72 Ni = 35 Nc = 37	+				
Fiol-DeRoque et al., Spain(34)	Self-conducted (online application)	Health care workers	PsyCovidApp, targeting emotional skills, healthy lifestyle behaviour, burnout, and social support	N = 436 Ni = 248 Nc = 234	NoAssoc	NoAssoc	NoAssoc	NoAssoc	
Guan, et al. (Study 2B), China(44)	Self-conducted (online)	General population	Self-compassion writing task	N = 79 Ni = 40 Nc = 39	+				
Guan et al. (Study 3), China(44)	Self-conducted (online)	College students	Self-compassion meditation and writing tasks	N = 95 Ni = 50 Nc = 45	+				
Kahlon et al., US(40)	Trained callers (Telephone-based)	Homebound older adults and people with disabilities	Phone call with empathetic conversations	N = 240 Nc = 120 Ni = 120	+		+		
Khademi et al., Iran)27(	Self-conducted	COVID-19 patients	Performed mandala colouring	N = 70 Ni = 35 Nc = 35	+				
Liu et al., China(48)	Trainer	COVID-19 patients	Jacobson's relaxation techniques	Intervention = 51 Ni = 25 Nc = 26	+			+	
Malboeuf-Hurtubise et al., Canada)41(	Trained students in psychology (Video)	School students	Mindfulness-based interventions and philosophical discussions on COVID-19	N = 37 (Pre-post intervention assessment)	+				

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Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Significance				
					Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Parizad et al., Iran(28)	Nurses	COVID-19 Patients	Guided imagery session with five audio tracks	N = 110 Nc = 55 Ni = 55	+				
Shaygan et al., Iran)29(	Self-conducted (online multimedia)	COVID-19 patients	Cognitive behavioural techniques, stress management, mindfulness, and positive psychotherapy	N = 48 Ni = 26 Nc = 22		+			
Smith et al., US)42(	Self-conducted (online application)	Obstetric and gynaecology patients	Mobile meditation app: encouragement, mindfulness meditation, sleep stories, nature sounds	N = 101 Ni = 50 Nc = 51	+	+	+	+	
Solianik et al., Lithuania)38(	Tai chi instructor (online)	General population >60 years	Tai chi practice	N = 30 Ni = 15 Nc = 15	+	+	+		
Wadhen et al., UK(39)	Yoga teacher (online)	General population	Yoga class via zoom meetings	N = 34 Nc = 17 Ni = 17	NoAssoc	NoAssoc	+		
Zhang et al., China(50)	Self-conducted (Online)	Residents	Mindfulness-Based Stress Reduction program	N = 51 Ni = 29 Nc = 28	+	+	+		

Note: + denotes Statistically significant association with post-intervention mental health improvements.
NoAssoc denotes No association between measure and post-intervention symptoms.
± denotes mixed results of association and no association, due to difference in timing of outcome measurement
Grey shade denotes not applicable.

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246 Table 4 Characteristics of the reviewed Non-randomised controlled study (NRS) studies, intervention, measurement and psychological outcome

Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Guan, et al. (Study 2A), China(44)	Self-conducted (online)	Young adults	A live session of self-compassion meditation	N = 64 (Pre-post intervention assessment)	+				
Giordano et al., Italy(30)	Trained music therapist	Clinical staff	Music therapy supplemented by guided imagery and music	N = 34 (Pre-post ntervention assessment)					+
González-García et al., Spain(35)	Mindfulness teacher and psychologist (online)	University students	Minilectures, mindfulness and compassion practices and self-reflection exercises	N = 66 (Pre-post intervention assessment)	+	+	+	+	
Gorbeña, et al., Spain(36)	Facilitators (interview and music playlists)	University students	Well-being and Personal Development Program	N = 151 Ni = 112 Nc= 39		+		+	
Kim et al., South Korea(45)	Psychiatrists (Telephone-based)	COVID-19 patients	30-minute telephone consultation providing information and support for COVID-19 infection and hospital isolation	Week1 n=33 Week2 n=19 (Pre-post intervention assessment)	±		±	±	±
Li et al., China(46)	Teacher (Video)	Nurses	Simulation training about using personal protective devices	N = 60 (Pre-post intervention assessment)	+		+		
Liu et al., China(47)	Self-conducted	Nurses	Diaphragmatic breathing relaxation training	N = 140 (Pre-post intervention assessment)	+		NoAssoc	+	
Luo et al., China(49)	Nurses	COVID-19 patients	Auricular point pressure	N = 84 Ni = 66 Nc = 18	+			+	

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Author	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Anxiety	Stress	Depres-sion	Insomnia / sleep quality	Suicidal idea
Matiz et al., Italy(31)	Socio-health educator and psychologist	School teachers	Mindfulness-oriented meditation training program	N = 67 (Pre-post intervention assessment)	+		+		
Niu et al., China(51)	Nurses	(Suspected) COVID-19	Providing information about isolation, treatment, and social emotional support	N = 137 (Pre-post intervention assessment)	+		+		
Philip et al., France(32)	Self-conducted (online application)	General population	KANOPEE app with virtual agent interaction	N = 47 (Pre-post intervention assessment)				+	
Riva et al. Italy)37(	Self-conducted	General population	360◦ VR video entitled 'The Secret Garden' and a series of social exercises	N = 38 (Pre-post intervention assessment)	NoAssoc	+	+		
Sylvia et al., US(43)	Psychologists, physicians, social workers, nurses (online)	Health care workers	Stress management and resilience training, relaxation response program	N = 102 (Pre-post intervention assessment)	+		+		
Tunuguntla et al., Global level(26)	Self-conducted (online)	General population	Yoga and meditation with sound therapy	N = 820 (Pre-post intervention assessment)				+	
Yang et al., China(52)	Psychotherapist and nurse (in-person and online)	COVID-19 patients	Psychotherapy, empathy, muscle and breath relaxation, and cognitive behavioural therapy	N = 35 (Pre-post intervention assessment)	+		+	+	

Note: + denotes Statistically significant association with post-intervention mental health improvements.
NoAssoc denotes No association between measure and post-intervention symptoms.
± denotes mixed results of association and no association, due to difference in timing of outcome measurement
Grey shade denotes not applicable.

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Of 24 anxiety interventions from 18 publications, 21 (87.5%) significantly lowered anxiety in intervention groups; 11 were RCT(27,28,33,38,40–42,44,48,50) and 10 quasi-experimental studies(31,35,43–47,49,51,52). Among the 11 RCT studies, five interventions (45.5%) were mindfulness and meditation-based(38,41,42,44,50). Two studies (18.2%) applied specific techniques such as Emotional Freedom Techniques (33) and progressive muscle relaxation and deep breathing techniques(48).

Fourteen studies provided different interventions to reduce depression, of which 11 (78.6%) significantly reduced depression(31,37–40,42,43,46,50–52). Of 14 studies, five of them (35.7%) were RCT; four (28.6%) related to mindfulness and meditation-based stress reduction(42,50), focusing on mind-body exercises including yoga(39) and tai chi(38) and one (7.1%) provided empathetic conversations through phone calls(40).

Of nine studies on insomnia, seven (77.8%) found that the interventions significantly reduced insomnia or improved sleep quality(26,32,42,47–49,52). Of these nine studies, only two (22.2%) were RCT, with one (11.1%) using a mobile meditation application(42), and the other using progressive muscle relaxation and deep breathing techniques(48).

Of total nine interventions against stress, seven (77.8%) had shown significant positive outcomes in stress reduction (29,35–38,42,50). One study (11.1%) shows no association of interventions and outcome(39), while one (11.1%) study showed positive and negative results due to timing of assessment(45). Of total 29 studies, three studies (10.3%) reported the effectiveness of interventions on self-compassion; participants reported significantly higher perception of self-compassion than those in the control group(35,43,44). One study of 29 studies (3.4%) using pre- and post- assessments show improvement of loneliness(40). Only one study (3.4%) examined suicide, 30-minute telephone consultation was provided which was effective after one week.)(45.

Risk of bias

The quality of reporting for 14 RCT studies was assessed using the Cochrane risk of bias tool. Table 5 shows the risk of bias by six domains. These RCT studies had no risk of attrition bias (incomplete outcome data) or reporting bias (selective reporting). Of 14 studies, five (35.7%) had low risk of bias for all six dimensions reflecting high quality RCT. Most studies had unclear reporting on the risk of allocation concealment, random sequence generation and blinding participants and personnel(27,38,41,44,48). Two studies (14.3%) reported high risk of bias on blinding of participants and personnel(39,50).

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Table 5 Risk of bias assessments for all 14 RCT studies

	Random sequence generation	Allocation concealment	Blinding of participants and personnel	Incomplete outcome data	Selective reporting	Other bias	Number of low risk out of six criteria
Dincer and Inangil (33)	Low	Low	Low	Low	Low	Low	6
Fiol-DeRoque et al.(34)	Low	Low	Low	Low	Low	Low	6
Guan et al. (study 2b)(44)	Unclear	Unclear	Unclear	Low	Low	Low	3
Guan et al. (study 3)(44)	Unclear	Unclear	Unclear	Low	Low	Low	3
Kahlon et al.(40)	Low	Low	Low	Low	Low	Low	6
Khademi et al.)27(	Low	Unclear	Unclear	Low	Low	Low	4
Liu et al.(48)	Unclear	Unclear	Unclear	Low	Low	Low	3
Malboeuf-Hurtubise et al.)4:	Unclear	Unclear	Low	Low	Low	Unclear	3
Parizad et al.(28)	Low	Low	Low	Low	Low	Low	6
Shaygan, et al.)29(	Low	Low	Low	Low	Low	Low	6
Smith, et al.)38,42(	Low	Low	Unclear	Low	Low	Low	5
Solianik, et al.)38(	Unclear	Unclear	Low	Low	Low	Low	4
Wadhen, et al.(39)	Low	Unclear	High	Low	Low	Low	4
Zhang, et al.(50)	Low	Unclear	High	Low	Low	Low	4

DISCUSSION

Study characteristics

All 29 studies in this review were carried out in 2020 in all geographical continents except Africa and Australia. The majority of studies (25, 86.2%) were from high- and upper-middle- income countries.Therefore, the varieties of interventions in this review can be useful for adaptive application in different country contexts, although there is no study from low-income countries. Psychological impacts are caused not only by direct consequences of the pandemic(53–55), but also by public health and social containment measures used by governments, notably movement restrictions and lockdowns(56). Over a half (19, 65.5%). of the psychological interventions reviewed in this study were delivered to COVID-19 patients, health care workers and general population via an online platform, as on-site intervention was not possible in the pandemic context. The psychological consequences from this pandemic are diverse, requiring a wide range of interventions and tools for measuring subsequent psychological outcomes, including suicidal ideation, depression, anxiety, stress, and insomnia.

Each outcome was evaluated using a variety of tools. The BDI was the most commonly used to assess suicidal ideation, followed by the HADS for screening depression, the STAI for screening anxiety, the PSS for screening stress, and the ISI for screening insomnia. The measures were all used for screening rather than diagnosis. Several measures were validated, with Cronbach's alpha values greater or equal to 0.7 indicating high reliability . The majority of the tools were self-assessment questionnaires that participants in the study could complete using the sample scoring system. However, no articles reported on the sensitivity and specificity of each measure**.**

Interventions and psychological outcomes

Overall, most interventions have positive impact in mitigating psychological consequences in COVID-19 patients, health care workers and general population. A single intervention can have combined effects on multiple psychological outcomes; for example one RCT using mobile meditation application including verbal and written encouragement, mindfulness meditation, sleep stories,

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nature sounds has significant positive effects on reduction of depression, anxiety, stress and insomnia(42). However, another study using an online application targeting emotional skills, healthy lifestyle behaviour, burnout, and social support reported no significant difference in the same measured outcomes(34).

Almost all 14 RCT studies that included mindfulness and meditation interventions to reduce depression, anxiety and stress showed significantly improved outcomes (depression(38–40,42,50), anxiety(27,28,33,38–42,44,50), stress(38,42,50)) with the exception of one study (39), which did not improve anxiety. Stress, anxiety, and depression is caused by sympathetic nervous system activation due to widespread depolarization throughout the brain and body; on the other hand, meditation and deep breathing lead to parasympathetic nervous system activation due to widespread inhibition and hyperpolarization(57). Recently, mindfulness and meditation-based interventions have increasingly been integrated into mental health interventions(58); meditation practices facilitate training of the self-regulation of attention and awareness, thereby enhancing control of mental processes, and consequently increasing wellbeing. Meditation and breathing techniques such as yoga, tai-chi, targeting body and mind are also applied and are effective in reducing stress, anxiety, and depression(59,60). Yoga, for example, helps with intellectual and mental exercise, improves the psychological management and monitoring of stress and negative emotions, and supports mental balance(60–62). Mindfulness and meditation-based interventions are readily available, and do not pose threats or risks. It can be applied as a first-line or complementary intervention for stress, anxiety, depression, and some emotional disorders during COVID-19.

Moreover, combining mindfulness intervention with online application technology can benefit people who are quarantined or in isolation wards with no direct access to psychotherapy. In this review, online multimedia psychoeducational interventions were demonstrated to significantly reduce stress(29); mindfulness-based stress reduction, cognitive–behavioural techniques, stress management techniques, and positive psychotherapy have been delivered via WhatsApp. Nowadays, internet-based, self-administrated multimedia and education can promote access to mental health interventions, particularly amidst COVID-19. Online-based interventions enable people to social distance and be time-efficient and cost-effective.

The pandemic does not allow in-person on-site intervention, causing a major shift to remote care(63–65) and telemedicine; the use of electronic communication to deliver health care services at a distance, becomes a key supporting measure for health services delivery systems during this pandemic(66,67). Several channels of telemedicine are accessible by patients and health professionals including telephone calls, video calls, interactive mobile health applications (m-Health), short message service, e-mail, secure remote patient monitoring, and video conferencing. Video calls via social media is the most common channel of communication, but special communication softwares (e.g., Zoom) or current messenger applications (e.g., WhatsApp, and WeChat) are popular and convenient options as well(68).

Furthermore, several countries have developed digital health technology to facilitate health care delivery through an online platform(69,70). Several challenges remain; online interventions cannot replace in-person therapy, notably when there is high risk of suicide which necessitates the presence of a psychotherapist or human interaction(71,72). Furthermore, several online counsellors(73) raised concerns from their experiences, such as a lack of non-verbal language for interpreting patients who struggle to verbally communicate their feelings(74) or significantly shorter counseling periods through the telephone due to “more superficial responses to questions”(75). However, the pandemic has raised awareness for the necessity of the online interventions where limitations can be resolved, as several studies have reported benefits of online services(76–79).

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Bias assessment

Most common reason across items of bias assessment was related to incomplete or unclear reporting in the RCT (80). Between 2011-2014, 24.7 % of the RCTs included in the Cochrane Review demonstrated an unclear risk of bias in the domain of incomplete outcome data (81). But all 14 RCT in the review presented complete outcome data and reported loss to follow-up for risk of attrition bias. In this review, each article was assessed by at least two authors and we compared the risk of bias assessment for each item in terms of 'high', 'low' or 'unclear' risk of bias between the two reviews.

More than half of the studies had unclear risk on allocation concealment due to the method of concealment was not described or inadequate detail to allow a clear judgment. Similarly, 50.1% of reported the April 2011 issue of the Cochrane Database of Systematic Reviews had unclear risk of bias on allocation concealment (82). Concealment of randomized allocation prevents an influence of patient characteristics on allocation to intervention and control groups. One study found that lack of adequate allocation concealment was associated with overestimation of treatment effect (83).

Two studies reported high risk of bias on blinding of participants and personnel (39,50). Blinding of participants and personnel prevents differences in patient management between groups and blinding of outcome assessors also prevents knowledge of the assigned intervention group influencing outcome measurement. However, both studies were conducted through online platform and the group memberships were not blind.

Limitations

Causal interpretations are limited by 15 non-randomised control study design. In addition to variation in sample size across studies, there were also variations in the mode of delivery of interventions, participant characteristics, and measurement tools across different study sites. Furthermore, a majority of the trials included small sample sizes which is commonplace for studies on psychological interventions. High heterogeneity of selected studies led to inability to conduct meta-analysis. There was also a lack of assessment of co-interventions, which provide external sources of support to the participants in the intervention and control groups and may have influenced results of the outcomes of an intervention. Lastly, post-intervention outcomes from the studies were measured from short follow-up durations, so it is not possible to conclude the long-term psychological effects of those interventions

CONCLUSION AND POLICY IMPLICATIONS

Of the total reviewed 29 studies, 26 diagnostic or screening measures were applied; eight for anxiety (30.9%), seven for depression (26.9%), five for stress (19.2%), five for insomnia (19.2%) and one for suicide (3.8%). Most studies conducted in high income and upper-middle income countries. Half of studies targeted general population and the remaining in COVID-19 patients and health care workers. Among 14 RCT studies, most interventions were effective in reducing psychological consequences.

Despite study limitations, the use of measurement measures and the strength of evidence from this review suggests an overall positive impact of the different interventions in terms of reducing psychological consequences in the population and health care workers. These effective interventions can be replicated and applied in other country settings, for which modes of delivery can be adjusted in line with country resources, epidemic situation, and social and health systems contexts. Moreover, it is necessary to improve the use of evidence in psychological interventions in mental health services.

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We recommend that these effective interventions can be replicated and applied in other country settings, for which modes of delivery can be adjusted in line with country resources, epidemic situation, and social and health systems contexts. In the first quarter of 2022, the pandemic is still far from over due to low immunization coverage in African countries and continued emergence of Variants of Concerns in the context of pressure to resuming the economic activities and opening the country borders (84). In ending the acute phase of pandemic, rapid scale up and promote vaccine acceptance are key policy actions (85).

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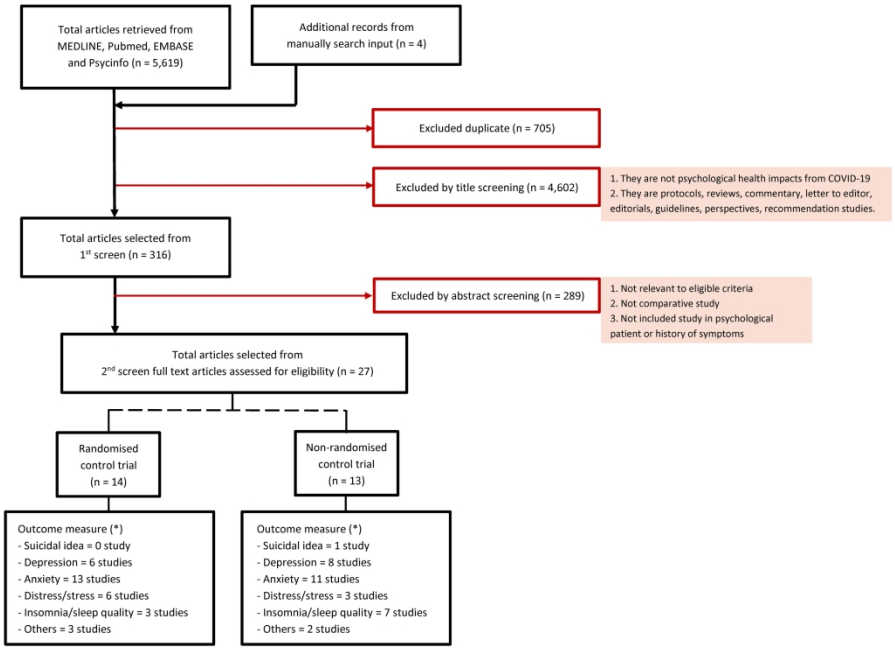
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Annex 1: Search term and results

1) PUBMED = 1,564
- COVID* OR novel corona* OR nCoV* OR 2019nCoV OR severe acute respiratory syndrome coronavirus 2 OR severe acute respiratory pneumonia outbreak OR SARS CoV 2 OR SARS-CoV-2 (206,274)
- therap*[MeSH Terms] OR intervention[MeSH Terms] OR consultation[MeSH Terms] (7,081,376)
- Mental Health [MeSH terms] OR Mental Disorders [MeSH terms] OR Mental Illness [MeSH terms] OR Suicide OR Depression OR Anxiety OR Psychological Distress [MeSH terms] OR Emotional Distress [MeSH terms] OR Insomnia [MeSH terms] (1,846,124)

2) EMBASE = 1,114
- COVID* (sh) or novel corona*.mp. or nCoV*.mp. or 2019nCoV.mp. or severe acute respiratory syndrome coronavirus 2 (sh) or severe acute respiratory pneumonia outbreak (sh) or SARS CoV 2.mp. or SARS-CoV-2.mp. (81,488)
- (treatment or therap*).mp. or intervention (sh) or consultation (sh) (13,205,742)
- (Mental Health or Mental Disorders or Mental Illness).mp. or Suicide (sh) or Depression (sh) or Anxiety (sh) or Psychological Distress.mp. or Emotional Distress.mp. or Insomnia (sh) (1,017,858)

3) PsycInfo = 1,524
- DE "Coronavirus" OR DE "COVID-19" OR DE "Severe Acute Respiratory Syndrome" (6,545)
- intervention or program or counseling (1,095,058)

4) MEDLINE(OVID) = 1,417
- (COVID* or novel corona* or nCoV* or 2019nCoV or severe acute respiratory syndrome coronavirus 2 or severe acute respiratory pneumonia outbreak or SARS CoV 2 or SARS-CoV-2).mp. (124,422)
- (therap* or therap* or intervention or consultation).mp. (6,269,680)
- (Mental Health or Mental Disorders or Mental Illness or Suicid* or Depression or Anxiety or Psychological Distress or Emotional Distress or Insomnia).mp. (638,321)

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Annex 2

Table A2 Tools for the assessment of psychological consequences with score classification

Outcomes	Tools	Method description	Score Classification	Studies
1. Suicidal Ideation Assessment	1.1 Beck Depression Inventory (BDI) (item 9)	A 21-item, self-reported rating inventory that measures attitudes and symptoms of depression. Item-9 of the test asked the potential to kill oneself (49).	Score 1-10: Normal Score 11-16: Mild Mood Disturbance Score 17-20: Borderline Clinical Depression Score 21-30: Moderate Depression Score 31-40: Severe Depression Score over 40: Extreme Depression	Kim et al., 2020(44)
2. Depression Assessment	2.1 Patient Health Questionnaire-9 (PHQ-9)	A 9-item assessment that is used to assess the frequency the patients experience depressive symptoms according to DSM-IV depression criteria(50).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score 15-19: Moderately Severe Score 20-27: Severe	Niu et al., 2021(50) Yang et al., 2020(51)
	2.2 Patient Health Questionnaire for Depression-8 (PHQ-8)	An 8-item assessment that consists of 8 of the 9 criteria on which the DSM-IV diagnosis of depressive disorders is based (51).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score 15-19: Moderately Severe Score 20-24: Severe	Kahlon et al., 2021(39)
	2.3 Patient Health Questionnaire-4 (PHQ-4)	A 4-item questionnaire that aims to allow for brief and accurate measurement of core symptoms or signs of depression and anxiety (52).	Score 0-2: Normal Score 3-5: Mild Score 6-8: Moderate Score 9-12: Severe	Sylvia et al., 2021(42)
	2.4 Depression Anxiety Stress Scale (DASS-21)	A 21-item test, short version of the original self-report questionnaire (DASS-42 item) to evaluate depression, anxiety, and stress symptoms with 7 items per subscale; (1) DASS-21 Depression, (2) DASS-21 Anxiety, and (3) DASS-21 Stress (53).	**Depression** Score 0-9: Normal Score 10-13: Mild Score 14-20: Moderate Score 21-27: Severe Score over 28: Extremely Severe **Anxiety** Score 0-7: Normal Score 8-9: Mild Score 10-14: Moderate Score 15-19: Severe	Riva et al., 2021(36), Wadhen et al., 2021(38), Fiol-DeRoque et al., 2021(33)

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Outcomes	Tools	Method description	Score Classification	Studies
			Score over 20: Extremely Severe **Stress** Score 0-14: Normal Score 15-18: Mild Score 19-25: Moderate Score 26-33: Severe Score over 34: Extremely Severe	
	2.5 Hospital Anxiety and Depression Scale (HADS)	A 14-item assessment used to measure anxiety and depression symptoms in medical patients. The HADS produces two scales with 7 items in each scale; one for anxiety (HADS–A) and one for depression (HADS–D) (54).	**Depression and Anxiety** Score 0-7: Normal Score 8-10: Borderline abnormal (borderline case) Score 11-21: Abnormal	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (HADS-D)(44)
	2.6 Self-rating Depression Scale (SDS)	A 20-item assessment to assess the subjective symptoms of depression for up to one week(55).	Score <50: Normal Score 50-59: Mild Depression Score 60-69: Moderate to Marked Major Depression Score >70: Extreme Major Depression	Liu et al., 2021(46), Li et al., 2020(45)
	2.7 Brief Symptom Inventory 18 (BSI-18)	An 18-item questionnaire, taken from the 53-item Brief Symptom Inventory (BSI). The scale includes three subscales (somatization, depression, and anxiety) (56).	The interpretation of the BSI-18 uses Global Severity Index (GSI) of distress represents sum across the 3 subscales range from 0-72. Higher scores indicate higher levels of psychological distress.	Zhang et al., 2021(49)
3. Anxiety assessment	3.1 Spielberger State-Trait Anxiety Inventory (STAI)	A 40-item self-report tool consisting of two parts for measuring state and trait anxiety with 20 questions in each section(57).	Score 20-31: Mild Score 32-42: Mild to Moderate Score 43-53: Moderate to Severe Score 54-64: Relatively Severe Score 65-75: Severe Score ≥ 76: Very Severe	Parizad et al., 2021(27), Khademi et al., 2021(26), González-García et al., 2021(34), Liu et al., 2020(47), Dincer et al., 2021(32), Guan et al., 2021(43)

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Outcomes	Tools	Method description	Score Classification	Studies
	3.2 Anxiety - Generalized Anxiety Disorder scale (GAD-7)	A 7-item questionnaire to detect symptoms of generalized anxiety disorder based on seven core symptoms and assess the frequencies with which respondents experience these symptoms within the prior 2 weeks (58).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score >=15: Severe	Kahlon et al., 2021(39), Luo et al., 2021(48), Niu et al., 2021(50), Yang et al., 2020(51)
	3.3 Self- Rating Anxiety Scale (SAS)	A 20-item assessment evaluating a variety of anxiety symptoms, including both psychological and somatic symptoms, that respondents experience within the past month (59).	Score 0-4: Normal Score 5-9: Mild Score 10-14: Moderate Score >=15: Severe	Liu et al., 2021 (46), Guan et al., 2021(43), Li et al., 2020(45)
	3.4 Behavior Assessment Scale for Children-3rd edition, (BASC III)	A test to identify and manage behavioral and emotional strengths and weaknesses of children and adolescents from three perspectives: teacher, parent and self (starting at age 8) (60).	Interpretation using the mean score. Higher scores indicate higher levels of anxiety.	Malboeuf-Hurtubise et al., 2021(40)
	3.5 Patient Health Questionnaire-4 (PHQ-4)	See 2.3		Sylvia et al., 2021 (42)
	3.6 Depression Anxiety Stress Scale (DASS-21)	See 2.4		Riva et al., 2021(36), Fiol-DeRoque et al., 2021(33), Wadhen et al., 2021(38)
	3.7 Hospital Anxiety and Depression Scale (HADS)	See 2.5	Score 20-44: Normal Score 45-59: Mild to Moderate Anxiety Score 60-74: Marked to Severe Anxiety levels	Solianik et al., 2021(37), Smith et al., 2021(41), Matiz et al., 2021(30), Kim et al., 2020 (44)
	3.8 Brief Symptom Inventory 18 (BSI-18)	See 2.7		Zhang et al., 2021 (49)

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Outcomes	Tools	Method description	Score Classification	Studies
4. Psychological distress and stress assessment	5. The General Health Questionnaire (GHQ-12)	A 12-item test, self-administered to evaluate mental distress and common mental disorders. There were multiple versions of GHQ including GHQ-60, GHQ-30, GHQ-28, and GHQ-12 (61).	The average sum of its items provides a scalar indicator of the degree of mental distress. Score of 11 or 12: typical Score >15: Suggesting evidence of distress Score >20: Severe problems with psychological distress	Gorbeña et al., 2021 (35)
	4.1 Perceived stress 14-item (PSS-14)	A 14-item scale used as a psychological instrument to measure the perception of stress, with seven positive items and seven negative items (62).	Score 0-18: Low Stress Score 19-37: Moderate Stress Score 38-56: High Stress	Wadhen et al., 2021(38), Shaygan et al., 2021 (28)
	4.2 Perceived Stress Scale 10-item (PSS-10)	A 10-item questionnaire about feelings and thoughts during the last month(63) .	Score 0-13: Low Stress Score 14-26: Moderate Stress Score 27-40: High Perceived Stress	Riva et al., 2021(36), Solianik et al., 2021(37), Smith et al., 2021(41), González-García et al., 2021(34)
	4.3 Depression Anxiety Stress Scale (DASS-21)	See 2.4	Score 0-13: Low Stress Score 14-26: Moderate Stress Score 27-40: High Perceived Stress	Fiol-DeRoque et al., 2021(33)
	4.4 Brief Symptom Inventory 18 (BSI-18)	See 2.7		Zhang et al., 2021 (49)
5. Insomnia assessment	5.1 Insomnia Severity Index (ISI)	A 7-item assessment to assess current sleep characteristics(64).	Score 0-7: No clinically significant insomnia Score 8-14: Subthreshold Insomnia Score 15-21: Clinically Significant Insomnia (Moderate) Score 22-28: clinically Significant Insomnia (Severe)	Tunuguntla et al., 2021(25), Philip et al., 2020(31), Fiol-DeRoque et al., 2021(33), Kim, et al., 2020(44)
	5.2 St. Mary's Hospital Sleep Questionnaire (SMH)	A 14-item test asking about patient's sleep experience and comprises items to demonstrable reliability, with some open-ended questions(65).	The test needs to be evaluated by experts.	Luo et al., 2021 (48)

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Outcomes	Tools	Method description	Score Classification	Studies
	5.3 PROMIS (Patient-Reported Outcomes Measurement Information System) Sleep Disturbance Short Form	Items focus on perceptions of sleep quality, sleep depth, and restoration associated with sleep, perceived difficulties with getting to sleep or staying asleep, and perceptions of the adequacy of and satisfaction with sleep (66).	Score <55: None to slight Score 55-59.9: Mild Score 60-69.9: Moderate Score >70: Severe	Smith et al., 2021 (41)
	5.4 Pittsburgh Sleep Quality Index (PSQI)	An 18-item index measuring seven sleep dimensions (67).	The overall PSQI score ranges from 0 to 21, with higher scores reflecting poorer sleep quality. A revised PSQI score of 7 or higher is considered poor sleep quality. If the score of any individual dimension is higher than 1, there is a problem with that specific dimension.	Liu et al., 2021(46), Yang et al., 2020(51)
	5.5 Sleep State Self-Rating Scale (SRSS)	A 10-item assessment used to assess sleep state.	SRSS over or equal to 23 points is considered to have sleep problems.	Liu et al., 2020(47)

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Annex 3

Table A3 Characteristics of the reviewed studies, intervention, measurement and psychological outcome

Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Dincer and Inangil, Turkey(32)	RCT	Certified emotional freedom technique	Nurses	Activating and deactivating signals to brain by stimulating points on skin	N = 72 Ni = 35 Nc = 37	Anxiety	STAI (+)
Fiol-DeRoque et al., Spain(33)	RCT	Self-conducted (online application)	Health care workers	PsyCovidApp, targeting emotional skills, healthy lifestyle behaviour, burnout, and social support	N = 436 Ni = 248 Nc = 234	Depression Anxiety Stress Insomnia	PHQ-8 (na) GAD7 (na) DASS-21 (na) ISI (na)
Guan, et al. (Study 2B), China(43)	RCT		General population	Self-compas sion writing task	N = 79 Ni = 40 Nc = 39	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Guan et al. (Study 3), China(43)	RCT		College students	Self-compassion meditation and writing tasks	N = 95 Ni = 50 Nc = 45	Anxiety Self-compassion	PHQ-4 (+) SCS (+)
Kahlon et al., US(39)	RCT	Trained callers (Telephone-based)	Homebound older adults and people with disabilities	Phone call with empathetic conversations	N = 240 Nc = 120 Ni = 120	Depression Anxiety Loneliness	GAD7 (+) GAD7 (+) UCLA Loneliness Scale (+), De Jong Loneliness Scale (na)
Khademi et al., Iran(26)	RCT	Self-conducted	COVID-19 patients	Performed mandala colouring	N = 70 Ni = 35 Nc = 35	Anxiety	GAD-7 (+)
Liu et al., China(47)	RCT	Trainer	COVID-19 patients	Jacobson's relaxation techniques	Intervention = 51 Ni = 25	Anxiety Insomnia	STAI (+) SRSS (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					Nc = 26		
Malboeuf-Hurtubise et al., Canada(40)	RCT	Trained students in psychology (Video)	School students	Mindfulness-based interventions and philosophical discussions on COVID-19	N = 37 (Pre-post intervention assessment)	Anxiety	BASC III (+)
Parizad et al., Iran(27)	RCT	Nurses	COVID-19 Patients	Guided imagery session with five audio tracks	N = 110 Nc = 55 Ni = 55	Anxiety	STAI (+)
Shaygan et al., Iran(28)	RCT	Self-conducted (online multimedia)	COVID-19 patients	Cognitive behavioural techniques, stress management, mindfulness, and positive psychotherapy	N = 48 Ni = 26 Nc = 22	Stress	PSS (+)
Smith et al., US(41)	RCT	Self-conducted (online application)	Obstetric and gynaecology patients	Mobile meditation app: encouragement, mindfulness meditation, sleep stories, nature sounds	N = 101 Ni = 50 Nc = 51	Depression Anxiety Stress Insomnia	HADS (+) HADS (+) PSS (+) PROMIS (+)
Solianik et al., Lithuania(37)	RCT	Tai chi instructor (online)	General population >60 years	Tai chi practice	N = 30 Ni = 15 Nc = 15	Depression Anxiety Stress	HADS (+) HADS (+) PSS-10 (+)
Wadhen et al., UK(38)	RCT	Yoga teacher (online)	General population	Yoga class via zoom meetings	N = 34 Nc = 17 Ni = 17	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (na), PSS (na)
Zhang et al., China(49)	RCT	Self-conducted (Online)	Residents	Mindfulness-Based Stress Reduction program	N = 51 Ni = 29	Depression Anxiety	BSI-18 (+) BSI-18 (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					Nc = 28	Stress	BSI-18 (+)
Kim et al., South Korea(44)	NRS	Psychiatrists (Telephone-based)	COVID-19 patients	Ward telephone providing information and support for COVID-19 infection and hospital isolation	Week1 n=33 Week2 n=19 (Pre-post intervention assessment)	Suicidal idea Depression Anxiety Insomnia	BDI: week1 (+), week2 (na) HADS: week1 (+), week2 (na HADS: week 1 (+), week 2 (na) ISI: week 1 (+), week 2 (na)
Guan, et al. (Study 2A), China(43)	NRS	Self-conducted (online)	Young adults	A live session of self-compassion meditation	N = 64 (Pre-post intervention assessment)	Anxiety	PHQ-4 (+)
Giordano et al., Italy(29)	NRS	Trained music therapist	Clinical staff	Music therapy supplemented by guided imagery and music	N = 34 (Pre-post intervention assessment)	Others: fear/ worry	MTC (+)
González-García et al., Spain(34)	NRS	Mindfulness teacher and psychologist (online)	University students	Minilectures, mindfulness and compassion practices and self-reflection exercises	N = 66 (Pre-post intervention assessment)	Anxiety Stress Self-compassion	DASS-21 (+) DASS-21 (+) DASS-21 (+)
Gorbeña, et al., Spain(35)	NRS	Facilitators (interview and music playlists)	University students	Well-being and Personal Development Program	N = 151 Ni = 112 Nc= 39	Stress	PSS (+)
Li et al., China(45)	NRS	Teacher (Video)	Nurses	Simulation training about using personal protective devices	N = 60 (Pre-post intervention assessment)	Depression Anxiety	SDS (+) SAS (+)
Liu et al., China(46)	NRS	Self-conducted	Nurses	Diaphragmatic breathing relaxation training	N = 140 (Pre-post	Depression	SDS (na)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
					intervention assessment)	Anxiety Insomnia	SAS (+) PSQI (+)
Luo et al., China(48)	NRS	Nurses	COVID-19 patients	Auricular point pressure	N = 84 Ni = 66 Nc = 18	Anxiety Insomnia	GAD-7 (+) SMH score (+)
Matiz et al., Italy(30)	NRS	Socio-health educator and psychologist	School teachers	Mindfulness-oriented meditation training program	N = 67 (Pre-post intervention assessment)	Depression Anxiety	HADS (+) HADS (+)
Niu et al., China(50)	NRS	Nurses	(Suspected) COVID-19	Providing information about isolation, treatment, and social emotional support	N = 137 (Pre-post intervention assessment)	Depression Anxiety	PHQ-9 (+) GAD-7 (+)
Philip et al., France(31)	NRS	Self-conducted (online application)	General population	KANOPEE app with virtual agent interaction	N = 47 (Pre-post intervention assessment)	Insomnia	ISI (+)
Riva et al. Italy(36)	NRS	Self-conducted	General population	360◦ VR video entitled 'The Secret Garden' and a series of social exercises	N = 38 (Pre-post intervention assessment)	Depression Anxiety Stress	DASS-21 (+) DASS-21 (na) DASS-21 (+), PSS (+)
Sylvia et al., US(42)	NRS	Psychologists, physicians, social workers, nurses (online)	Health care workers	Stress management and resilience training, relaxation response program	N = 102 (Pre-post intervention assessment)	Depression Anxiety Self-compassion	PHQ-4 (+) PHQ-4 (+) SCS (+)

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Author	Study Design	Provider (channel)	Target population	Intervention	N = Total sample Nc: Control, Ni: intervention)	Psychological outcome	Measurement and findings
Tunuguntla et al., Global level(25)	NRS	Self-conducted (online)	General population	Yoga and meditation with sound therapy	N = 820 (Pre-post intervention assessment)	Insomnia	ISI (+)
Yang et al., China(51)	NRS	Psychotherapist and nurse (in-person and online)	COVID-19 patients	Psychotherapy, empathy, muscle and breath relaxation, and cognitive behavioural therapy	N = 35 (Pre-post intervention assessment)	Depression Anxiety Insomnia	PHQ-9 (+) GAD-7 (+) PSQI (+)

RCT: Randomised controlled trial

NRS: Non-randomised controlled study

+ = Statistically significant association with post-intervention mental health improvements.

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	P1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	P1
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	P4, Line 73-81
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	P5, Line 87-90
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Table 1
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	P 5, Line 100
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Figure 1
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	P 5, Line 106
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Table 1
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	P9, Line 135
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	P9, Line 124
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Table 1
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	NA
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	NA
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	P9, Line 124

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	P9, Line 141
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	P9, Line 141, Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 6
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 4 and 5
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table 4
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	NA
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Table 6
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	P27, Line 329
	23b	Discuss any limitations of the evidence included in the review.	P27, Line 380
	23c	Discuss any limitations of the review processes used.	P27, Line 380
	23d	Discuss implications of the results for practice, policy, and future research.	P27, Line 390
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Not registered
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	NA
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	P30, Line 402

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
Competing interests	26	Declare any competing interests of review authors.	NA
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71
For more information, visit: http://www.prisma-statement.org/

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